

SOC Threat Detection in a Segmented Healthcare Network

(MediCore Health Systems)

BY

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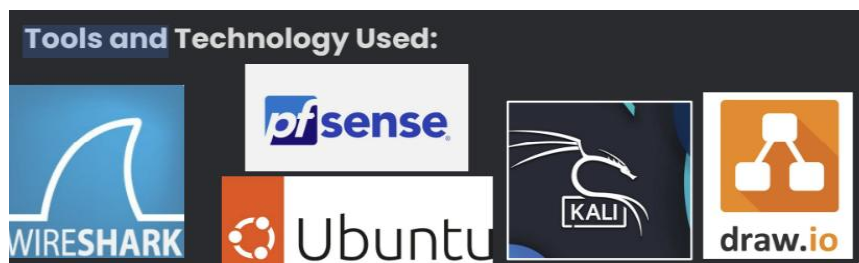


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LIST OF ACRONYMS AND ABBREVIATIONS

CIA: Confidentiality, Integrity, and Availability

DHCP: Dynamic Host Configuration Protocol

DMZ: Demilitarized Zone

DNS: Domain Name System

EHR: Electronic Health Record

FIM: File Integrity Monitoring

HTTPS: Hypertext Transfer Protocol Secure

ICMP: Internet Control Message Protocol

IoC: Indicator of Compromise

IoMT: Internet of Medical Things

IP: Internet Protocol

LAN: Local Area Network

MFA: Multi-Factor Authentication

NAT: Network Address Translation

SIEM: Security Information and Event Management

SMB: Server Message Block

SNMP: Simple Network Management Protocol

SOC: Security Operations Center

SSH: Secure Shell

TCP: Transmission Control Protocol

VPN: Virtual Private Network

WAN: Wide Area Network

CHAPTER ONE

1.0 EXECUTIVE SUMMARY

MediCore Health Systems, during a routine security assessment of their system, observed repeated port scanning and authentication failures from an internal network targeting their EHR and new Patient Portal. Due to recent expansion of their digital healthcare services, including telehealth and Internet of Medical Things (IoMT) devices, protecting patient's private information, and ensuring uninterrupted service is of great importance for the smooth running of the organization and to ensure compliance. The SOC team was tasked with identifying the source, containing, identifying the source, containing the threat, and strengthening security controls to protect patient data.

To begin with, the methodology utilized to achieve this commenced with the mapping out of the existing enterprise network using Draw.io and planning a segmented network by creating security zones; WAN, LAN (for clinical systems, administrative computers, medical devices, Internet of Medical things (IOMT), and security operation centers), and DMZ hosts (EHR and Patient telehealth portal). In addition, pfsense was implemented to enable the deployment of firewalls to control the traffic within the different security zones. Furthermore, the Ubuntu Server with the hosted EHR and patient portal was created in the DMZ zone while Wazuh was used for centralizing log management, correlation, and monitoring of the security events. Wireshark captured the network traffic, and Kali Linux was used to simulate reconnaissance and authentication attacks.

The objective of this assessment is to detect the present threat and block it. Then, evaluate the current network architecture, create a network that can identify and limit security weaknesses, monitor and detect malicious intent, and respond appropriately.

Final suggestions to strengthen the organization's cybersecurity posture were made after several important indications of compromise (IoCs) were found and addressed.

CHAPTER TWO

2.0 PROJECT INTRODUCTION

MediCore Health Systems is an expanding healthcare company providing critical digital health solutions, which include Electronic Health Records (EHR), patient portals, telemedicine services, medical billing, insurance processing, and IoMT (Internet of Medical Things) devices management. The company processes and stores a lot of confidential information about patients; therefore, its network is an appealing target for cybersecurity attacks. In order to increase accessibility of its services, MediCore launched a DMZ-based patient portal allowing patients to access their healthcare services and records using the internet.

During normal security monitoring, it was discovered that there is some abnormal network activity associated with regular port scanning and authentication failures targeting critical network components such as the EHR environment and the patient portal. Besides, suspicious traffic coming from the internal workstations was noticed, indicating possible cases of unauthorized access and lateral movement through the network, which poses a challenge for the current security design. Such issues pointed out the necessity for enhanced network security controls, improved visibility into the network activity and minimized risks of data leaks, system downtime and non-compliance with regulatory requirements.

To tackle these issues, this SOC (Security Operations Center) Capstone project was carried out to assess the security posture of the organization within a simulated enterprise healthcare setting. This capstone project showcases how an enterprise network architecture can be described, segmented, and secured using the modern standard security technologies of the industry. The assessment involves designing a segmented network via Draw.io, setting up firewall policies via pfSense, configuring central monitoring via Wazuh, monitoring the network traffic via Wireshark, and conducting simulated cyberattacks via Kali Linux. The main aim of this project is to show how the overlapping of security controls, monitoring, and response to incidents can help to ensure security of patient information.

CHAPTER THREE

3.0 Methodology

The project followed a structured, a six-stage approach to design, secure, monitor, test, and evaluate MediCore Health Systems' simulated healthcare network.

3.1 Network Design and Segmentation

The network architecture was outlined in Draw.io to highlight the key systems, security zones, communications pathways, and trust boundaries. The architecture differentiated the security zones; WAN, LAN, and DMZ. Clinical workstations, administrative workstations, SOC workstations, Internet of medical things (IOMT), Security incident and event management (SIEM-wazuh) and Kali (simulated attacker network) were mapped in the LAN. The EHR, patient portal and telehealth server was mapped in the DMZ as this is public-facing, and more prone to security attacks. Then, the IP address was planned and documented for each network zone (192.168.10.0/24 for WAN, 192.168.20.0/24 for LAN, 192.168.30.0/24 for DMZ)

3.2 pfSense Configuration and Firewall Policies

pfSense was used as the main firewall and router between all network zones. The WAN network was created on the pfsense using the already documented IP address, while enabling DHCP. The first adapter (WAN interface) was enabled on NAT network, then used with an external machine to initialize and configure the pfSense before the LAN and DMZ security zones adapters were set under internal network on the pfSense. Interfaces were defined and configured for WAN, LAN, and DMZ. Furthermore, the DHCP server was enabled and configured to automatically assign IP addresses to any machine we add in each security zone (LAN and DMZ). Then, the machines were added into their respective zones under internal network.

Furthermore, the firewall rules were set. They utilized a principle of least privilege and default deny rule. The DMZ and LAN firewall rule was initially configured to allow any traffic. This is to simulate the impact of the attack (hydra attack planned with attacker network; Kali) on the EHR and patient portal hosted in the DMZ. Later, the firewall rule was adjusted to only allow internet

access to the patient portal via the HTTPS protocol, while any access from the Internet to the EHR and other internal systems was not allowed. Clinical machines could access approved medical services, while other machines were prevented from unnecessary access to sensitive systems.

3.3 Wazuh Deployment and Log Collection

A **Wazuh Manager** was deployed on an Ubuntu server and critical systems to provide centralized security monitoring. Furthermore, wazuh agents were installed on Ubuntu server (hosting the EHR and Patient Telehealth services), and workstation systems. Logs were collected from the Ubuntu server and systems that had wazuh agents installed. The collection plan focused on failed logins, privilege changes, firewall denials, suspicious processes, file-integrity changes, and unusual access to critical systems.

3.4 Attack Simulation

An isolated **Kali Linux** system represented the attacker network. Testing was performed only within the controlled laboratory environment for ethical concerns. **Nmap** was used to conduct reconnaissance (host discovery, service identification, and TCP port scans against the patient portal and approved internal targets). Controlled authentication testing generated repeated failed login attempts against the Ubuntu-hosted service. Each activity was performed in stages, beginning with baseline traffic, followed by reconnaissance and authentication attempts, so that the resulting events could be clearly identified and compared.

3.5 Network Traffic Capture

Wireshark was used to capture traffic at selected observation points, including the Kali attacker interface, the target server interface, and network segments near the pfSense firewall. Captures were taken before, during, and after each simulation. Display filters were applied to isolate source and destination IP addresses, TCP connection attempts, reset packets, HTTP or HTTPS sessions, and repeated authentication-related communications. This approach provided packet-level evidence of both the attacks and the firewall's response.

3.6 Detection, Correlation, and Active Response

Wazuh alerts, pfSense logs, Ubuntu authentication records, and Wireshark packet captures were correlated using timestamps, source addresses, destination systems, ports, and event types. This helped distinguish ordinary activity from reconnaissance and repeated access attempts. When suspicious behaviour was confirmed, response actions included blocking the identified source through pfSense, restricting unnecessary services, reviewing affected accounts, strengthening authentication controls, and validating that the malicious traffic was no longer successful. Follow-up testing was then performed to confirm that the new controls improved detection and containment without interrupting authorized clinical activity.

CHAPTER FOUR

4.0 Phase-by-Phase Analysis

4.1 PHASE 1: Network Design & Segmentation (Draw.io & pfSense)

Objective

The objective of this phase was to design a secure enterprise network architecture that separates critical healthcare resources into distinct security zones and implements access controls to reduce the risk of unauthorized access and lateral movement. Proper network segmentation improves the confidentiality, integrity, and availability of Electronic Health Records (EHR), Internet of Medical Things (IoMT) devices, and other sensitive healthcare services.

4.1.1 Key Actions

4.1.1.1 Enterprise Network Design (Draw.io)

The existing healthcare network was documented using **Draw.io** to illustrate the organization's infrastructure and communication paths. The design included the Internet connection (WAN), pfSense firewall, internal LAN, and DMZ.

The network was segmented into the following security zones:

- ❖ **WAN** – Internet connection.
- ❖ **DMZ** – EHR server, Patient Portal and Telehealth Server
- ❖ **LAN** – Internal user network (user, admin, and security operations center (SOC) workstations, Kali attacker network, wazuh SIEM, and IoMT devices)

Each network segment was assigned its own subnet to improve traffic control and simplify security monitoring.

4.1.1.2 pfSense Configuration

pfSense was configured as the organization's perimeter firewall and gateway. The following interfaces were assigned and configured:

- ❖ WAN Interface
- ❖ LAN Interface
- ❖ DMZ Interface

Routing between network segments was configured through pfSense while ensuring that all traffic passed through the firewall before reaching protected resources. DHCP was enabled for the LAN and DMZ interfaces to automatically assign IP addresses to any workstation assigned to each network.

Interface configuration

The security zones adapters are configured to enable their assignment on pfsense. Fig 3.2 shows the adapters that were enabled (Medicore WAN on NAT Network, LAN, and DMZ on Internal network).

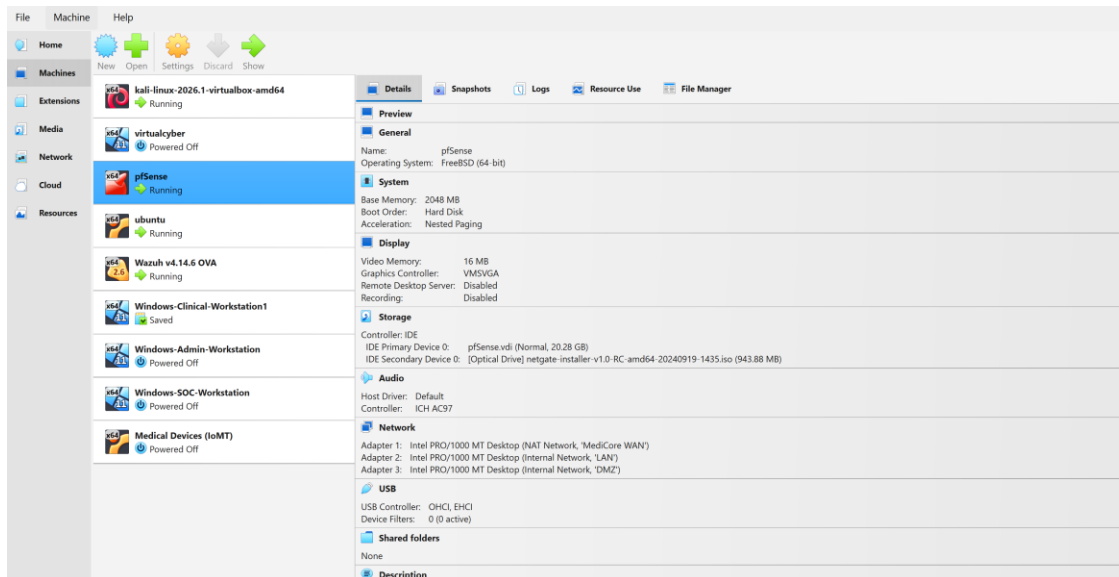


Fig 4.1: pfSense interface configuration showing the defined LAN, WAN, and DMZ adapters.

Interface assignment

Here, Interfaces are assigned and renamed to the appropriate security zones.

Direct Host configuration Protocol (DHCP) configuration.

This protocol automatically assigns IP addresses to any system added on a network. This is configured to automatically assist in assigning IP addresses to any systems or servers added in the LAN and DMZ subnets. Fig 2.6 and 2.7 shows the DHCP server configurations for the LAN and the DMZ. The range of the IP addresses are set in this portion, and the server automatically follows this in assigning the IP addresses.

The screenshot shows the pfSense web interface for the DHCP Server configuration of the LAN interface. The top navigation bar includes links for System, Interfaces, Firewall, Services, VPN, Status, Diagnostics, and Help. The breadcrumb trail is Services / DHCP Server / LAN. A yellow warning banner states: "ISC DHCP has reached end-of-life and will be removed in a future version of pfSense. Visit System > Advanced > Networking to switch DHCP backend." Below this, the "LAN" tab is selected. The configuration is divided into two main sections: "General Settings" and "Primary Address Pool".

General Settings	
DHCP Backend	ISC DHCP
Enable	☒ Enable DHCP server on LAN interface
BOOTP	☐ Ignore BOOTP queries
Deny Unknown Clients	Allow all clients When set to Allow all clients, any DHCP client will get an IP address within this scope/range on this interface. If set to Allow known clients from any interface, any DHCP client with a MAC address listed in a static mapping on <i>any</i> scope(s)/interface(s) will get an IP address. If set to Allow known clients from only this interface, only MAC addresses listed in static mappings on this interface will get an IP address within this scope/range.
Ignore Denied Clients	☐ Ignore denied clients rather than reject This option is not compatible with failover and cannot be enabled when a Failover Peer IP address is configured.
Ignore Client Identifiers	☐ Do not record a unique identifier (UID) in client lease data if present in the client DHCP request This option may be useful when a client can dual boot using different client identifiers but the same hardware (MAC) address. Note that the resulting server behavior violates the official DHCP specification.

Primary Address Pool	
Subnet	192.168.20.0/24
Subnet Range	192.168.20.1 - 192.168.20.254
Address Pool Range	From: 192.168.20.10 To: 192.168.20.100 The specified range for this pool must not be within the range configured on any other address pool for this interface.
Additional Pools	+ Add Address Pool If additional pools of addresses are needed inside of this subnet outside of the above range, they may be specified here.

Fig 4.2: DHCP server configuration for LAN on pfsense.

Services / DHCP Server / DMZ

ISC DHCP has reached end-of-life and will be removed in a future version of pfSense. Visit [System > Advanced > Networking](#) to switch DHCP backend.

LAN DMZ

General Settings

DHCP Backend	ISC DHCP
Enable	☒ Enable DHCP server on DMZ interface
BOOTP	☐ Ignore BOOTP queries
Deny Unknown Clients	☐ Allow all clients When set to Allow all clients, any DHCP client will get an IP address within this scope/range on this interface. If set to Allow known clients from any interface, any DHCP client with a MAC address listed in a static mapping on any scope(s)/interface(s) will get an IP address. If set to Allow known clients from only this interface, only MAC addresses listed in static mappings on this interface will get an IP address within this scope/range.
Ignore Denied Clients	☐ Ignore denied clients rather than reject This option is not compatible with failover and cannot be enabled when a Failover Peer IP address is configured.
Ignore Client Identifiers	☐ Do not record a unique identifier (UID) in client lease data if present in the client DHCP request This option may be useful when a client can dual boot using different client identifiers but the same hardware (MAC) address. Note that the resulting server behavior violates the official DHCP specification.

Primary Address Pool

Subnet	192.168.30.0/24
Subnet Range	192.168.30.1 - 192.168.30.254
Address Pool Range	192.168.30.10 From 192.168.30.100 To The specified range for this pool must not be within the range configured on any other address pool for this interface.
Additional Pools	+ Add Address Pool If additional pools of addresses are needed inside of this subnet outside the above range, they may be specified here.

Server Options

WINS Servers	WINS Server 1
	WINS Server 2

Fig 4.3: DHCP server configuration for DMZ on pfSense.

```
Starting CRON... done.
pfSense 2.8.1-RELEASE amd64 20260513-1943
Bootup complete

FreeBSD/amd64 (MedCore.home.arpa) (ttyv0)
VirtualBox Virtual Machine - Netgate Device ID: 78b16881aafda2a786a8

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on MedCore ***

WAN (wan) -> em0 -> v4/DHCP4: 192.168.10.3/24
LAN (lan) -> em1 -> v4: 192.168.20.1/24
DMZ (opt1) -> em2 -> v4: 192.168.30.1/24

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address     11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults        13) Update from console
5) Reboot system                    14) Enable Secure Shell (sshd)
6) Halt system                      15) Restore recent configuration
7) Ping host                        16) Restart PHP-FPM
8) Shell

Enter an option: 
```

Fig 4.4: pfSense start-up page showing all the interface configured.

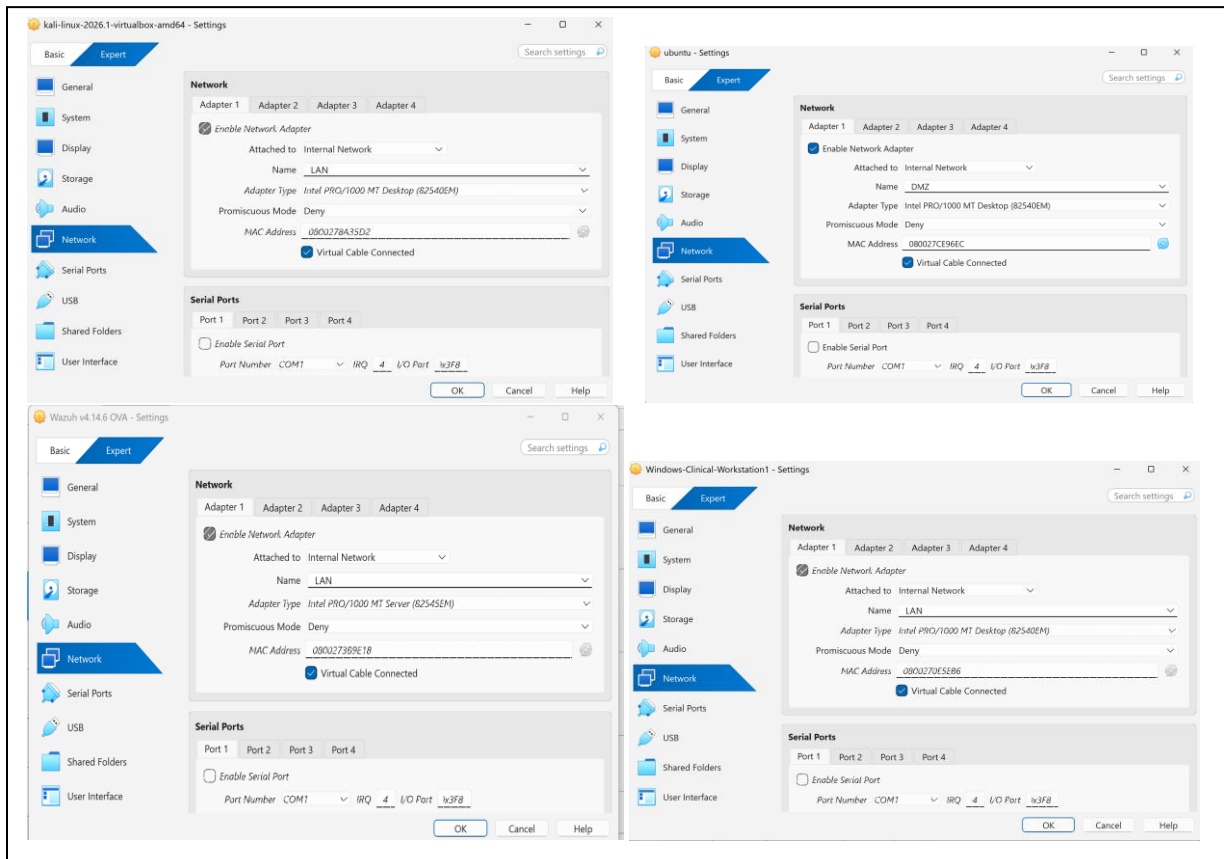


Fig 4.5: Asset assignment to their respective security zones

Firewall rule creation and Implementation.

Firewall rules were created for each subnet, LAN, and DMZ to filter traffic going from each network to another. This rule follows a principle of least privilege, allowing only business-required communications while blocking unnecessary access. The LAN subnet was initially set to be able to access any destination network to help simulate the impact of this kind of setting on the security of an organization if an attacker were to gain access to one of workstations on the LAN network. Examples of what the rule should do is listed below.

- ❖ Allow HTTPS (TCP 443) from the Internet to the Patient Portal located in the DMZ.
- ❖ Allow authorized clinical workstations to access the EHR server.
- ❖ Permit Wazuh agents to communicate with the Wazuh Manager.
- ❖ Allow DNS and DHCP traffic where required.

- ❖ Block direct Internet access to the EHR database.
- ❖ Block IoMT devices from initiating connections to administrative systems.
- ❖ Deny all unsolicited inbound traffic unless explicitly permitted.
- ❖ Log blocked and suspicious traffic for monitoring and forensic analysis.

Enabling simple network management protocol (SNMP).

SNMP is a protocol that manages all network devices including router, switches, firewall and servers, and most time, it is expected to only be configured for the administrator for management of services. It was initially enabled for all to be able to study the impact of this misconfiguration on the security of the organization.

The screenshot shows the pfSense web interface for configuring SNMP. The top navigation bar includes links for System, Interfaces, Firewall, Services, VPN, Status, Diagnostics, and Help. The main content area is titled 'Services / SNMP'. It is divided into several sections: 'SNMP Daemon' with an 'Enable' checkbox checked; 'SNMP Daemon Settings' with fields for 'Polling Port' (161), 'System Location', 'System Contact', and 'Read Community String' (public); 'SNMP Traps Enable' with an 'Enable' checkbox unchecked; 'SNMP Modules' with a list of modules (MibII, Netgraph, PF, Host Resources, UCD, Regex) all checked; and 'Interface Binding' with 'Internet Protocol' set to IPv4 and 'Bind Interfaces' set to All.

Fig 4.6: Enabling SNMP for all interfaces on pfSense.

Expected Security Controls

The following security controls were implemented during this phase:

- ❖ Enterprise network segmentation using dedicated security zones (LAN, WAN, DMZ)
- ❖ DMZ isolation for Internet-facing services
- ❖ Default-deny firewall policy.
- ❖ Least privilege access control
- ❖ Network Address Translation (NAT)
- ❖ Controlled routing between network segments
- ❖ Logging of firewall events
- ❖ Restricted communication between clinical, administrative, IoMT, and public networks
- ❖ Protection of Electronic Health Records (EHR) from direct Internet access

Observed Results / Notes

The segmented network was successfully implemented, and all network zones communicated only through pfSense. Firewall rules correctly restricted unauthorized communication between network segments while permitting approved business traffic. The patient portal remained accessible from the Internet through the DMZ, whereas the EHR server was protected from direct external access. Connectivity testing confirmed that blocked traffic was denied according to the configured firewall policies, and firewall logs recorded both permitted and denied connections for later security analysis.

Artifacts

Figure 4.7: Enterprise Network Diagram created using Draw.io

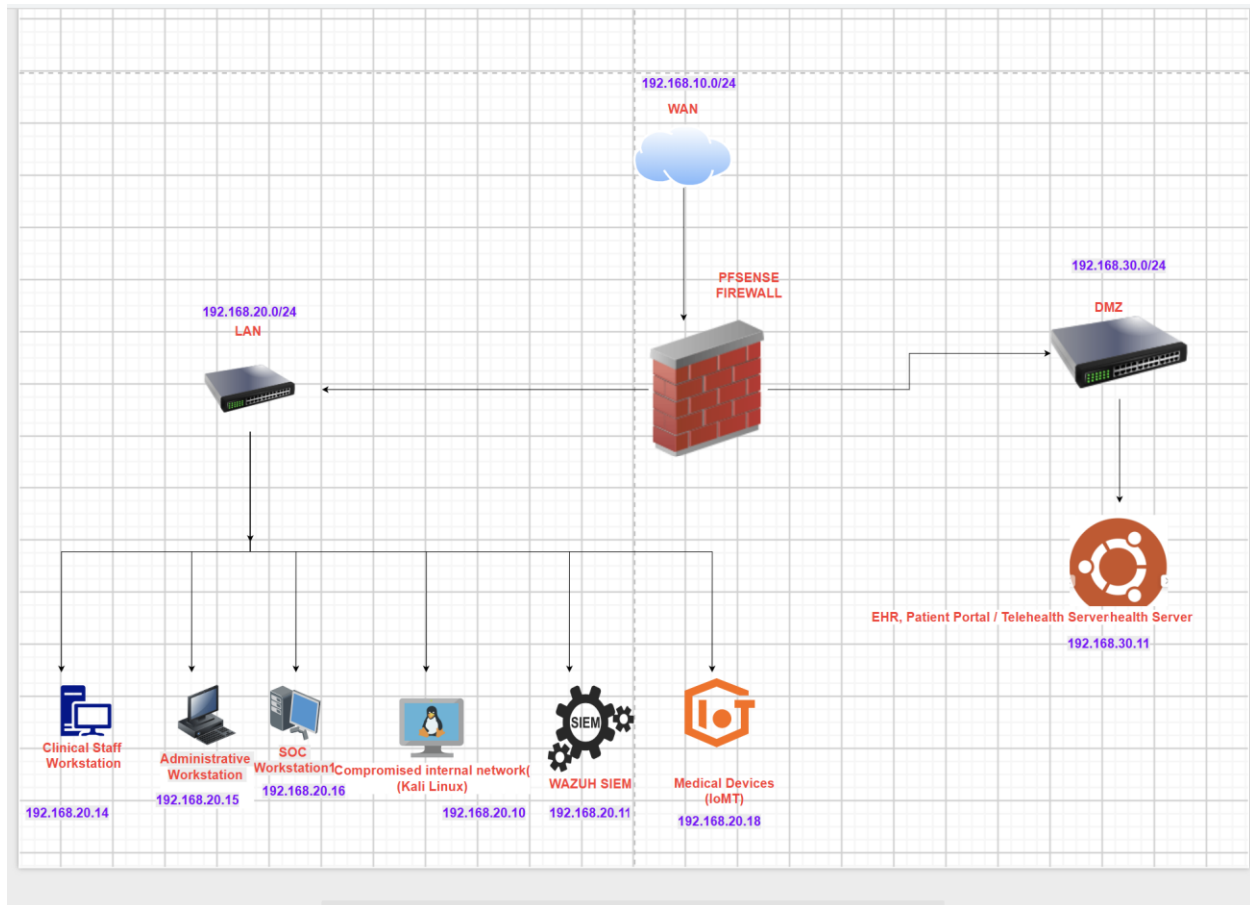


Fig: 4.8 **Interface configuration on pfsense**

The figure displays two screenshots of the pfSense web interface, specifically the 'Interfaces' section, showing the configuration for two different interfaces: LAN (em1) and DMZ (em2).

Top Screenshot: LAN (em1) Configuration

- General Configuration:**
 - Enable:** ☒ Enable interface
 - Description:** LAN
 - IPv4 Configuration Type:** Static IPv4
 - IPv6 Configuration Type:** None
 - MAC Address:** xxxxxxxxxxxxxx
 - MTU:** (blank)
 - MSS:** (blank)
 - Speed and Duplex:** Default (no preference, typically autoselect)
- Static IPv4 Configuration:**
 - IPv4 Address:** 192.168.20.1
 - IPv4 Upstream gateway:** None
- Reserved Networks:** (empty)

Bottom Screenshot: DMZ (em2) Configuration

- General Configuration:**
 - Enable:** ☒ Enable interface
 - Description:** DMZ
 - IPv4 Configuration Type:** Static IPv4
 - IPv6 Configuration Type:** None
 - MAC Address:** xxxxxxxxxxxxxx
 - MTU:** (blank)
 - MSS:** (blank)
 - Speed and Duplex:** Default (no preference, typically autoselect)

Figure 4.9: pfSense Firewall Rules

pfSense
COMMUNITY EDITION

System ▾

Interfaces ▾

Firewall ▾

Services ▾

VPN ▾

Status ▾

Diagnostics ▾

Help ▾

7

Firewall / Rules / Edit

Firewall Rule

Action

Pass

Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

LAN

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

Any

Choose which IP protocol this rule should match.

Source

Source

☐ Invert match

LAN subnets

Source Address

/

Destination

Destination

☐ Invert match

Any

Destination Address

/

Extra Options

Log

☐ Log packets that are handled by this rule

Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the Status: System Logs: Settings page).

Description

Default allow LAN to any rule

A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset label and displayed in the firewall log.

Advanced Options

Display Advanced

Rule Information

Tracking ID

0100000101

Updated

7/28/26 17:41:03 by admin@192.168.20.10 (Local Database)

Save

pfSense
COMMUNITY EDITION

System ▾

Interfaces ▾

Firewall ▾

Services ▾

VPN ▾

Status ▾

Diagnostics ▾

Help ▾

7

Firewall / Rules / Edit

Firewall Rule

Action

Pass

Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

DMZ

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

Any

Choose which IP protocol this rule should match.

Source

Source

☐ Invert match

DMZ subnets

Source Address

/

Destination

Destination

☐ Invert match

Any

Destination Address

/

Extra Options

Log

☐ Log packets that are handled by this rule

Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the Status: System Logs: Settings page).

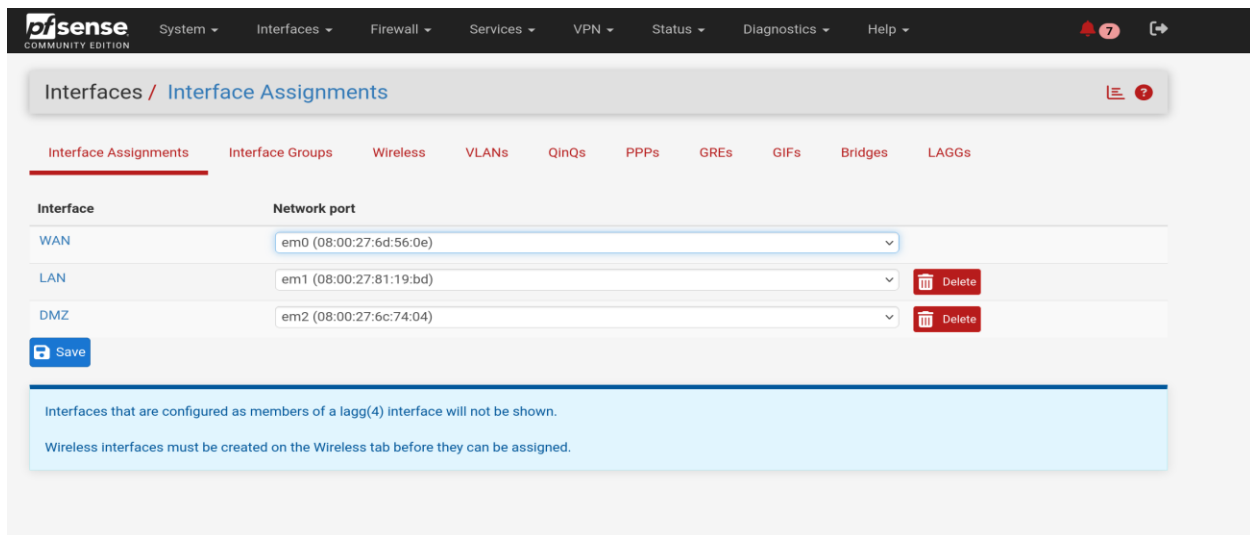
Description

A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset label and displayed in the firewall log.

Advanced Options

Display Advanced

Figure 4.10: pfSense Interface Assignments



4.2 PHASE 2: Attack Simulation & Traffic Capture (Kali Linux & Wireshark)

4.2.1 Network Traffic Capture Using Wireshark

Wireshark was deployed within the network to capture traffic generated during the attack simulations.

Packet captures were collected from:

- ❖ The Kali Linux attack workstation
- ❖ The Ubuntu patient portal server
- ❖ Network segments connected to the pfSense firewall.

Traffic captures were performed before, during, and after each simulated attack to establish baseline activity and identify abnormal network behavior.

Display filters were applied to isolate relevant traffic, including:

- ❖ TCP SYN packets
- ❖ SSH traffic (Port 22)
- ❖ HTTP and HTTPS traffic (Ports 80 and 443)

- ❖ ICMP packets
- ❖ TCP reset packets.
- ❖ Source and destination IP addresses associated with the attacker.

Traffic Analysis

Captured packets were analyzed to identify indicators of suspicious activity.

The analysis focused on:

- Repeated connection attempts from a single source
- Port scanning behaviour across multiple TCP ports
- Multiple failed SSH authentication attempts
- Firewall responses to blocked traffic.
- TCP session establishment and termination
- Communication between internal hosts and DMZ services

The captured evidence was later correlated with pfSense firewall logs and Wazuh alerts to validate attack detection and support incident investigation.

4.2.2 Observed Results / Notes

The simulated attacks successfully generated network traffic representative of reconnaissance and unauthorized ssh hydra attack. Wireshark captured the Nmap scans, repeated SSH connection attempts, and communication between the attacker and target systems (ubuntu server). Packet analysis confirmed multiple SYN packets directed toward the target host, service discovery requests, and repeated authentication attempts against the SSH service. The captured traffic provided valuable evidence for subsequent analysis within Wazuh and the pfSense firewall logs.

Artifacts

Figure 4.11: Kali Linux Nmap Port Scan

```
Session Actions Edit View Help
|
| pfsync0
|   Type: ilan Speed: 0 Kbps
|   Status: down
|   Traffic stats: 0.00 Kb sent, 0.00 Kb received
MAC Address: 08:00:27:81:19:BD (Oracle VirtualBox virtual NIC)

Nmap scan report for MediCore.home.arpa (192.168.20.1)
Host is up (0.00084s latency).

PORT      STATE SERVICE
161/udp   open  snmp
|
| snmp.interfaces:
|   em0
|     IP address: 192.168.10.3 Netmask: 255.255.255.0
|     MAC address: 08:00:27:6d:56:0e (Oracle VirtualBox virtual NIC)
|     Type: ethernetCsmacd Speed: 1 Gbps
|     Status: up
|     Traffic stats: 332.85 Kb sent, 818.89 Kb received
|   em1
|     IP address: 192.168.20.1 Netmask: 255.255.255.0
|     MAC address: 08:00:27:81:19:bd (Oracle VirtualBox virtual NIC)
|     Type: ethernetCsmacd Speed: 1 Gbps
|     Status: up
|     Traffic stats: 198.95 Kb sent, 805.43 Kb received
|   em2
|     IP address: 192.168.30.1 Netmask: 255.255.255.0
|     MAC address: 08:00:27:6c:74:04 (Oracle VirtualBox virtual NIC)
|     Type: ethernetCsmacd Speed: 1 Gbps
|     Status: up
|     Traffic stats: 80.44 Kb sent, 35.32 Kb received
|   enc0
|     Type: wwanPP2 Speed: 0 Kbps
|     Status: down
|     Traffic stats: 0.00 Kb sent, 0.00 Kb received
|   lo0
|     IP address: 127.0.0.1 Netmask: 255.0.0.0
|     Type: softwareLoopback Speed: 0 Kbps
|     Status: up
|     Traffic stats: 3.93 Kb sent, 3.93 Kb received
|   pflog0
|     Type: ifPwType Speed: 0 Kbps
|     Status: down
|     Traffic stats: 16.24 Kb sent, 0.00 Kb received
|   pfsync0
|     Type: ilan Speed: 0 Kbps
|     Status: down
|     Traffic stats: 0.00 Kb sent, 0.00 Kb received
MAC Address: 08:00:27:81:19:BD (Oracle VirtualBox virtual NIC)

Nmap done: 2 IP addresses (2 hosts up) scanned in 0.76 seconds

(kali@kali)~$
$ nmap 192.168.30.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-03 09:09 -0400
Nmap scan report for 192.168.30.1
Host is up (0.00075s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  https

Nmap scan report for 192.168.30.11
Host is up (0.0023s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh

Nmap done: 256 IP addresses (2 hosts up) scanned in 9.23 seconds
```

This figure shows the results of an Nmap network scan performed from the Kali Linux attacker (192.168.20.10) against the 192.168.30.0/24 subnet. The scan identified two active hosts within the target network and enumerated their exposed services. The healthcare gateway (192.168.30.1) was found to have ports 53 (DNS), 80 (HTTP), and 443 (HTTPS) open, while the Ubuntu server (192.168.30.11) was discovered with TCP port 22 (SSH) open. Identifying the SSH service enabled the attacker to select the Ubuntu server as the target for the subsequent SSH brute-force

attack using Hydra. This reconnaissance phase demonstrates how attackers gather information about network hosts and exposed services before attempting unauthorized access.

Table 4.2: nmap scan summary

Key evidence shown
Attacker: Kali Linux (192.168.20.10)
Target subnet: 192.168.30.0/24
Reconnaissance tool: Nmap
Active hosts discovered:
192.168.30.1 – Gateway/Firewall
192.168.30.11 – Ubuntu Server
Open ports identified:
192.168.30.1: TCP 53 (DNS), 80 (HTTP), 443 (HTTPS)
192.168.30.11: TCP 22 (SSH)
Purpose: To identify reachable hosts and vulnerable services prior to launching the SSH brute-force attack.

Figure 4.12: Kali Linux SSH Brute-Force Attack (Hydra)

```

kali@kali:~$ hydra -L users.txt -P password.txt -t 4 -V ssh://192.168.30.11
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-08-03 23:11:20
[DATA] max 4 tasks per 1 server, overall 4 tasks, 24 login tries (1:6/p:4), ~6 tries per task
[DATA] attacking ssh://192.168.30.11:22/
[ATTEMPT] target 192.168.30.11 - login "admin" - pass "password" - 1 of 24 [child 0] (0/0)
[ATTEMPT] target 192.168.30.11 - login "admin" - pass "ubuntu" - 2 of 24 [child 1] (0/0)
[ATTEMPT] target 192.168.30.11 - login "admin" - pass "123456789" - 3 of 24 [child 2] (0/0)
[ATTEMPT] target 192.168.30.11 - login "admin" - pass "10Analytics" - 4 of 24 [child 3] (0/0)
[22][ssh] host: 192.168.30.11 login: admin password: password
[ATTEMPT] target 192.168.30.11 - login "ubuntu" - pass "password" - 5 of 24 [child 0] (0/0)
[ATTEMPT] target 192.168.30.11 - login "ubuntu" - pass "ubuntu" - 6 of 24 [child 2] (0/0)
[ATTEMPT] target 192.168.30.11 - login "ubuntu" - pass "123456789" - 7 of 24 [child 1] (0/0)
[ATTEMPT] target 192.168.30.11 - login "ubuntu" - pass "10Analytics" - 8 of 24 [child 3] (0/0)
[22][ssh] host: 192.168.30.11 login: ubuntu password: ubuntu
[ATTEMPT] target 192.168.30.11 - login "medicore" - pass "password" - 9 of 24 [child 2] (0/0)
[ATTEMPT] target 192.168.30.11 - login "medicore" - pass "ubuntu" - 10 of 24 [child 0] (0/0)
[22][ssh] host: 192.168.30.11 login: medicore password: password
[ATTEMPT] target 192.168.30.11 - login "johnson" - pass "password" - 13 of 24 [child 2] (0/0)
[ATTEMPT] target 192.168.30.11 - login "johnson" - pass "ubuntu" - 14 of 24 [child 1] (0/0)
[ATTEMPT] target 192.168.30.11 - login "johnson" - pass "123456789" - 15 of 24 [child 3] (0/0)
[ATTEMPT] target 192.168.30.11 - login "johnson" - pass "10Analytics" - 16 of 24 [child 2] (0/0)
[ATTEMPT] target 192.168.30.11 - login "10Analytics" - pass "password" - 17 of 24 [child 0] (0/0)
[ATTEMPT] target 192.168.30.11 - login "10Analytics" - pass "ubuntu" - 18 of 24 [child 1] (0/0)
[ATTEMPT] target 192.168.30.11 - login "10Analytics" - pass "123456789" - 19 of 24 [child 3] (0/0)
[ATTEMPT] target 192.168.30.11 - login "10Analytics" - pass "10Analytics" - 20 of 24 [child 2] (0/0)
[ATTEMPT] target 192.168.30.11 - login "" - pass "password" - 21 of 24 [child 0] (0/0)
[ATTEMPT] target 192.168.30.11 - login "" - pass "ubuntu" - 22 of 24 [child 1] (0/0)
[ATTEMPT] target 192.168.30.11 - login "" - pass "123456789" - 23 of 24 [child 3] (0/0)
[ATTEMPT] target 192.168.30.11 - login "" - pass "10Analytics" - 24 of 24 [child 2] (0/0)
1 of 1 target successfully completed, 3 valid passwords found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-08-03 23:11:32

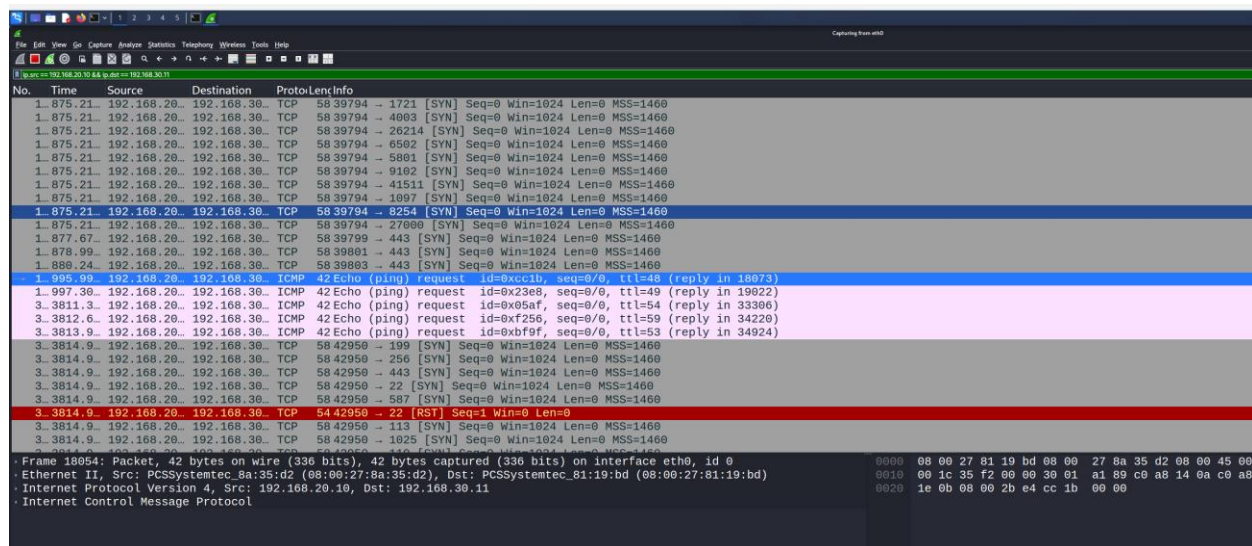
```

This figure shows the execution of an SSH brute-force attack from the Kali Linux attacker (192.168.20.10) against the Ubuntu server (192.168.30.11) using the THC Hydra password-cracking tool. Hydra systematically attempted multiple username and password combinations from the provided username (users.txt) and password (password.txt) wordlists against the SSH service on TCP port 22. The output displays each login attempt in real time and reports three valid credentials that successfully authenticated to the target server. This simulated attack demonstrates how weak or predictable passwords can be exploited to gain unauthorized access to critical healthcare systems. The attack generated corresponding network traffic captured in Wireshark and security alerts detected by the Wazuh SIEM, allowing the incident to be investigated and correlated across multiple security monitoring tools.

Table 4.3: Hydra attack summary

Key evidence shown
Attacker: Kali Linux (192.168.20.10)
Target: Ubuntu Server (192.168.30.11)
Attack tool: THC Hydra
Protocol: SSH (TCP port 22)
Attack type: SSH brute-force/password guessing
Evidence: Multiple login attempts using different username/password combinations
Outcome: Hydra reports "3 valid passwords found," confirming successful authentication with weak credentials

Figure 4.13: Wireshark Packet Capture – Port Scanning Activity

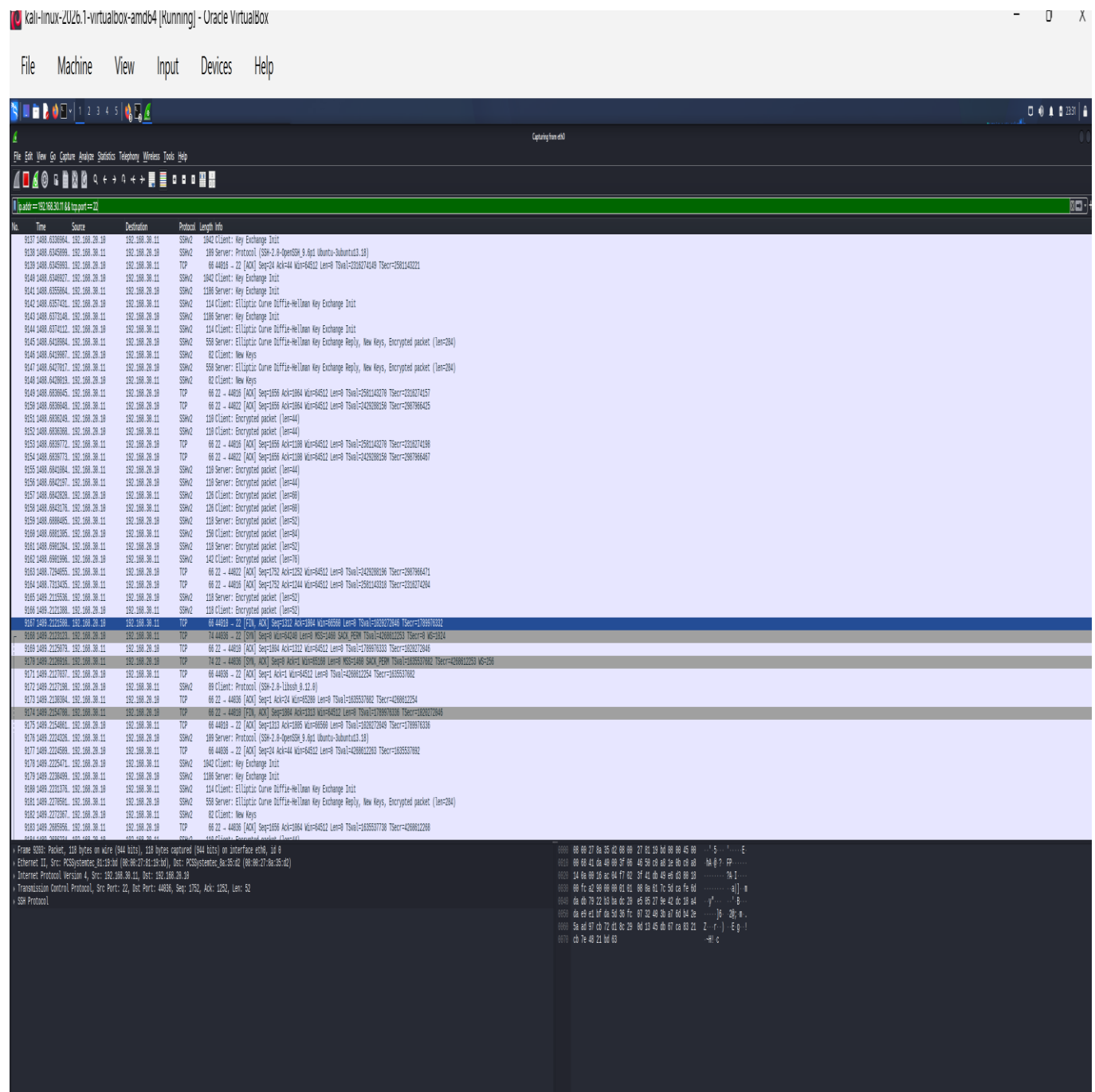


The image shows a Wireshark packet capture window with the following details:

- Filter:** 192.168.20.10 <-> 192.168.30.11
- Packet List:** A list of 24 packets. Packets 1-10 are TCP SYN packets from 192.168.20.10 to 192.168.30.11 on various ports (1721, 4063, 26214, 6502, 5801, 9102, 41511, 1097, 8254, 27000). Packets 11-13 are ICMP Echo (ping) requests from 192.168.30.10 to 192.168.30.11. Packets 14-16 are TCP SYN packets from 192.168.30.10 to 192.168.30.11 on ports 443, 443, and 443. Packets 17-19 are TCP SYN packets from 192.168.30.10 to 192.168.30.11 on ports 443, 443, and 443. Packets 20-22 are TCP SYN packets from 192.168.30.10 to 192.168.30.11 on ports 443, 443, and 443. Packets 23-24 are TCP SYN packets from 192.168.30.10 to 192.168.30.11 on ports 443, 443, and 443.
- Packet Details:** The selected packet (No. 19) is an ICMP Echo (ping) request. The details pane shows: Ethernet II, Src: PCSysmtec 8a:35:d2 (08:00:27:8a:35:d2), Dst: PCSysmtec 81:19:bd (08:00:27:81:19:bd); Internet Protocol Version 4, Src: 192.168.20.10, Dst: 192.168.30.11; Internet Control Message Protocol.
- Packet Bytes:** The packet bytes pane shows the raw data of the selected packet: 0000 08 00 27 81 19 bd 08 00 27 8a 35 d2 08 00 45 00 0010 00 1c 35 f2 00 00 30 01 a1 89 c9 a8 14 0a c0 a8 0020 1e 0b 08 00 2b e4 cc 1b 00 00

This Wireshark capture shows the initial reconnaissance traffic generated by the Kali Linux attacker workstation (192.168.20.10) targeting the Ubuntu server (192.168.30.11). The captured packets include multiple TCP SYN packets sent to various destination ports, ICMP Echo (ping) requests, and connection attempts to TCP port 22 (SSH). These packets indicate that the attacker was identifying active hosts, probing open services, and verifying network connectivity before initiating the SSH brute-force attack. The repeated SYN packets demonstrate automated port scanning activity, while the ICMP requests confirm that the target host was reachable. This traffic represents the reconnaissance phase of the cyber attack and provides evidence of malicious network activity prior to credential-based exploitation.

Figure 4.14: Wireshark Packet Capture – SSH Authentication Attempts



This Wireshark capture shows the SSH communication between the Kali Linux attacker (192.168.20.10) and the Ubuntu server (192.168.30.11) over TCP port 22. The capture displays the SSH protocol handshake, including Key Exchange Init, Elliptic Curve Diffie-Hellman (ECDH) key exchange, New Keys, and multiple encrypted SSH packets. These packets confirm that an SSH session was successfully established before authentication attempts were made. The encrypted packets indicate that all credentials and subsequent communication were protected by SSH encryption, preventing the actual usernames and passwords from being viewed in Wireshark. This traffic demonstrates the attacker's attempt to authenticate to the Ubuntu server, which was later correlated with Wazuh alerts generated during the SSH brute-force attack.

4.3 PHASE 3: Security Event Monitoring & Response (Wazuh SIEM)

Objective

The objective of this phase was to establish centralized security monitoring capable of collecting, correlating, and analyzing security events generated across the MediCore Health Systems network. Wazuh; Security Information and Event Management (SIEM) was deployed to improve visibility into network and endpoint activities, detect indicators of compromise, and support timely incident response. The monitoring environment was configured to identify suspicious activities such as repeated authentication failures, network reconnaissance, privilege escalation, and unauthorized access attempts while providing actionable alerts for security analysts.

Key Actions

1. Wazuh Manager and Agent Deployment

A **Wazuh Manager** was deployed within the secure LAN to function as the organization's centralized security monitoring platform. Wazuh agents were installed on selected Ubuntu servers and endpoint systems located within the LAN and DMZ to continuously collect security-related events.

The deployment included:

- ❖ Installation and configuration of the Wazuh Manager
- ❖ Registration of Wazuh agents with the Manager
- ❖ Verification of agent connectivity and health status
- ❖ Centralized collection of endpoint and operating system logs

This configuration enabled centralized visibility across multiple network segments and ensured that security events from critical systems were consolidated into a single monitoring platform.

2. Alert Configuration and Security Monitoring

Built-in Wazuh detection capabilities were configured to identify activities that could indicate attempted or successful cyberattacks.

Such alerts include.

- ❖ Multiple failed SSH authentication attempts
- ❖ Port scanning and reconnaissance activities
- ❖ Privilege escalation attempts.
- ❖ New user account creation
- ❖ Unauthorized changes to critical system files
- ❖ Suspicious process execution
- ❖ Firewall deny events.
- ❖ Unauthorized access to sensitive healthcare systems
- ❖ File Integrity Monitoring (FIM) alerts

Alert severity levels were used to prioritize incidents according to their potential impact on the organization's operations and patient data.

3. Active Response Configuration

To reduce response time and contain potential threats, Wazuh Active Response was configured for selected high-risk events. The active response was used to block any ssh future brute force attack.

The firewall wall block rule was also created on pfsense to block the LAN subnet from communicating with the DMZ. Furthermore, aliases pass rule were created for ip addresses (e.g the admin and other crucial personnel) that would need access to the DMZ.

Observed Results / Notes

The Wazuh deployment successfully collected security events from monitored endpoints(Ubuntu) and the pfsense firewall. During the simulated attacks, the SIEM generated alerts for repeated SSH authentication failures and network reconnaissance activities initiated from the Kali Linux attacker

machine. Security events were correlated using timestamps, source IP addresses, usernames, and event severity, allowing suspicious behaviour to be quickly identified and investigated. Active response policies (Active response on wazuh and block rule in pfsense) successfully blocked repeated unauthorized connection attempts, while the Wazuh dashboard provided centralized visibility into detected events and supported efficient incident analysis.

Artifacts

Figure 6.1: Wazuh Dashboard Overview

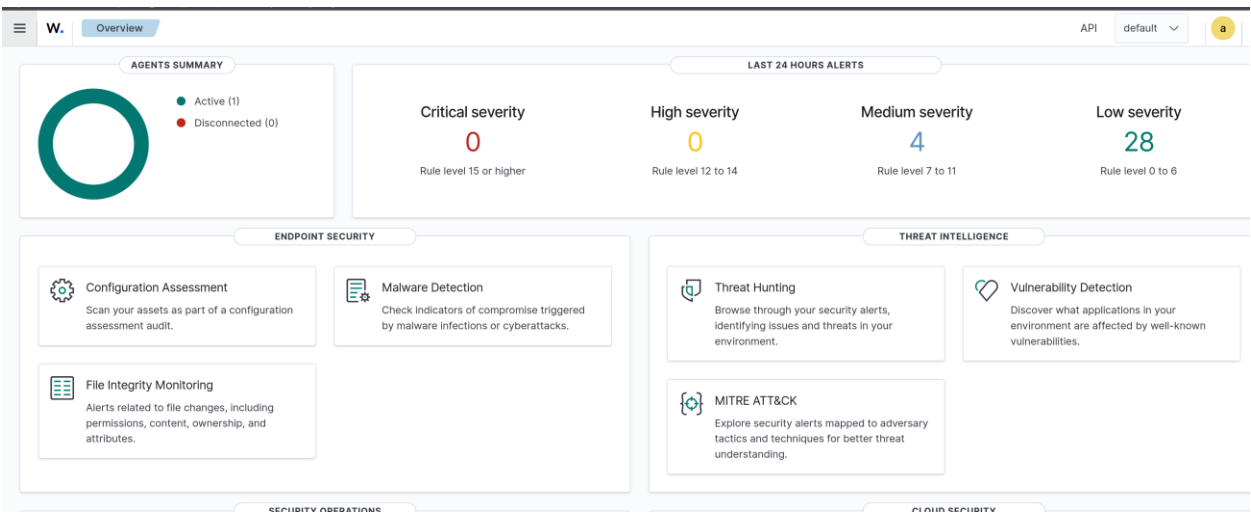


Figure 6.2: Registered Wazuh Agents

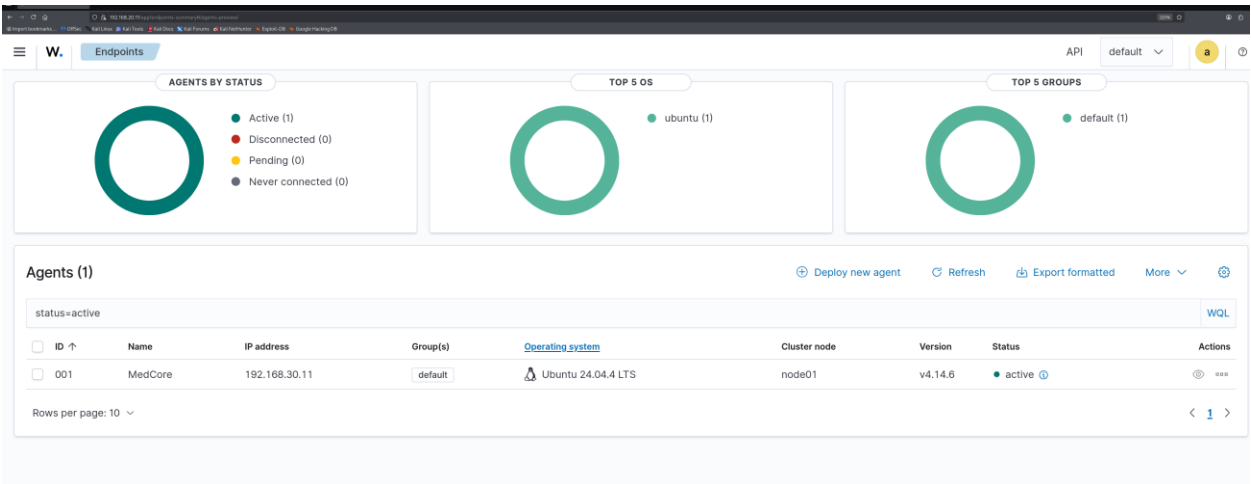


Figure 6.3: Authentication Failure Alerts

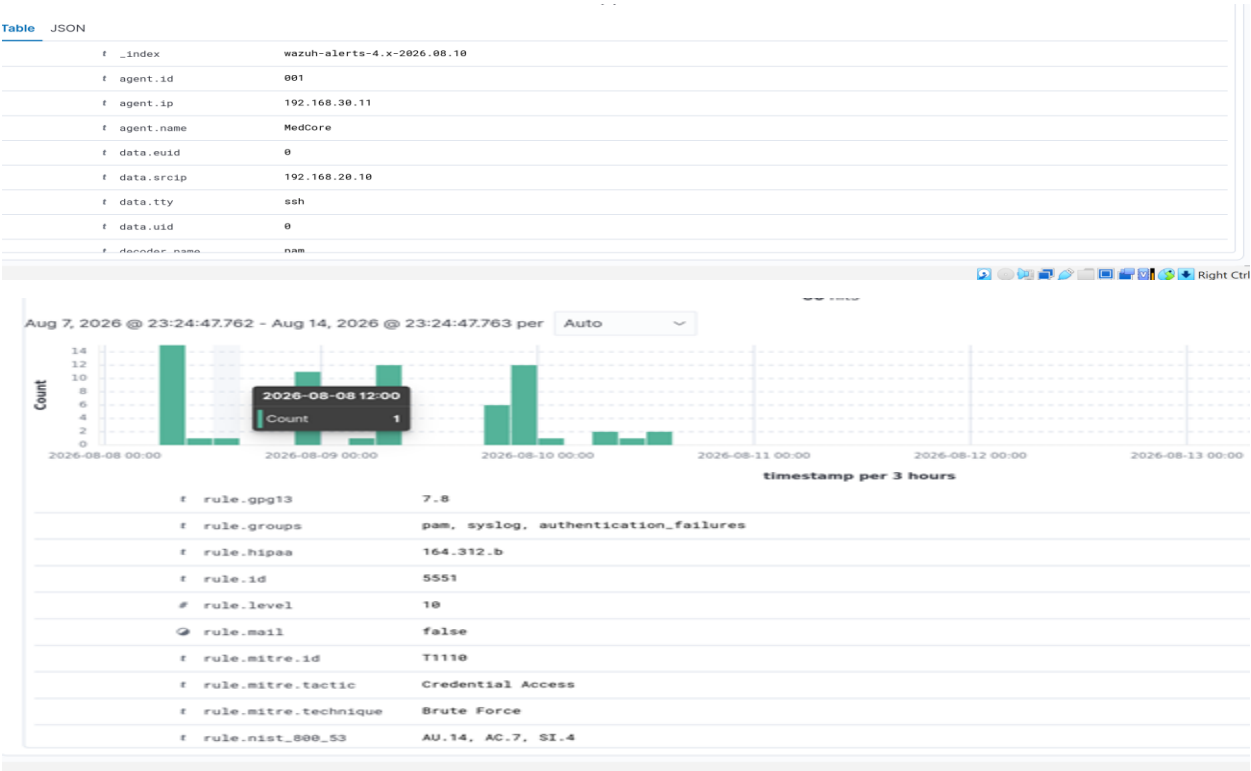
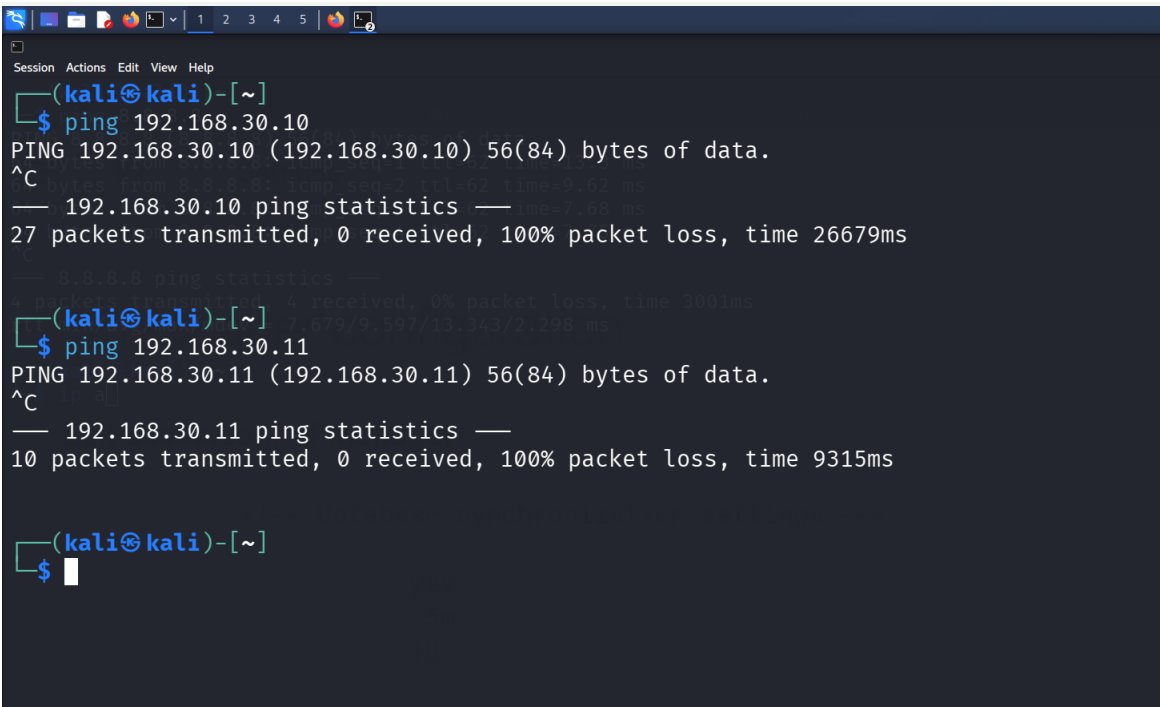


Figure 6.4: Active Response / Blocked IP Events



CHAPTER FIVE

5.0 Final Findings & Impact

5.1 Final Findings

The security assessment identified several weaknesses within the MediCore Health Systems network that could expose critical healthcare systems and sensitive patient information to cyber threats. The most significant issues were insufficient network segmentation, overly permissive firewall policies, and limited centralized monitoring prior to implementing the proposed security controls. These weaknesses increased the likelihood of unauthorized access, lateral movement between network segments, and delayed detection of malicious activities targeting the Electronic Health Record (EHR) system and patient portal.

The simulated attacks demonstrated how common reconnaissance and authentication-based attacks could be detected when multiple security tools operate together. Nmap port scans generated abnormal connection attempts that were visible in Wireshark packet captures, recorded by the pfSense firewall, and subsequently correlated as security events within Wazuh SIEM. Likewise, repeated SSH authentication attempts initiated from the Kali Linux attack workstation produced authentication failure events that were collected from the Ubuntu server, correlated by Wazuh, and presented as security alerts. The combination of packet captures, firewall logs, and endpoint security events provided a comprehensive view of each attack throughout its lifecycle.

The investigation demonstrated that implementing network segmentation and enforcing least-privilege firewall policies significantly reduced the attack surface. Restricting communication between the LAN, DMZ, EHR network, IoMT devices, and administrative systems limited unauthorized access and reduced opportunities for attackers to move laterally after compromising a host. Centralized monitoring also improved the Security Operations Center's ability to identify suspicious activities quickly and respond before significant damage could occur.

If these weaknesses remained unaddressed, the organization could face serious consequences affecting the confidentiality, integrity, and availability (CIA) of its information systems. Unauthorized access to Electronic Health Records could expose Protected Health Information (PHI), resulting in patient privacy violations and regulatory penalties. Successful modification of healthcare records could compromise the integrity of clinical information, potentially affecting patient treatment decisions. Additionally, attacks against critical servers or medical devices could disrupt healthcare services, reducing system availability and impacting patient care. Such incidents could also lead to non-compliance with healthcare regulations, including HIPAA, and damage the organization's reputation.

Several Indicators of Compromise (IoCs) were confirmed during the investigation, including:

Repeated TCP SYN packets consistent with Nmap port scanning.

Multiple failed SSH authentication attempts originating from the Kali Linux attack host.

High volumes of connection attempts from a single source IP address within a short period.

Firewall deny events generated by pfSense for unauthorized connection attempts.

Wazuh alerts identifying authentication failures and reconnaissance activities.

Correlated timestamps linking packet captures, firewall logs, and endpoint security events.

Source and destination IP addresses associated with suspicious network activity.

Overall, the findings demonstrate that combining Draw.io for network planning, pfSense for network segmentation and firewall enforcement, Wireshark for packet analysis, Kali Linux for controlled attack simulation, and Wazuh SIEM for centralized monitoring provides an effective layered security approach. These controls significantly improve threat detection, reduce the organization's attack surface, support regulatory compliance, and strengthen MediCore Health Systems' ability to protect patient information and maintain uninterrupted healthcare services.

CHAPTER SIX

6.0 Recommendations

Based on the findings of the security assessment, the following recommendations are proposed to strengthen MediCore Health Systems' cybersecurity posture. The recommendations are prioritized according to their potential impact on reducing organizational risk and improving the protection of Electronic Health Records (EHR), Internet of Medical Things (IoMT) devices, and other critical healthcare services.

6.1 Priority 1: Strengthen Technical Security Controls

The highest priority should be securing network access to critical healthcare systems.

Recommended actions include:

- ❖ Harden SSH by disabling direct root login and enforcing key-based authentication.
- ❖ Implement Multi-Factor Authentication (MFA) for administrative accounts, VPN access, and privileged users.
- ❖ Apply stricter pfSense firewall policies using a **default-deny** approach, allowing only approved business traffic.
- ❖ Restrict unnecessary inbound and outbound ports, particularly SSH (22), RDP (3389), SMB (445), and other unused services.
- ❖ Separate IoMT devices, clinical workstations, administrative systems, and public-facing services into dedicated network segments with limited inter-network communication.
- ❖ Disable unused services and remove unnecessary software from servers and endpoints.
- ❖ Apply secure configuration baselines to servers and network devices.

These measures reduce the attack surface and make unauthorized access significantly more difficult.

6.2 Priority 2: Improve Security Operations Processes

Security controls should be supported by well-defined operational procedures.

Recommended improvements include:

- ❖ Establish centralized log retention policies that retain security logs for at least 90–365 **days** to support investigations and compliance requirements.
- ❖ Develop documented incident response playbooks for common security events such as port scanning, brute-force attacks, malware infections, privilege escalation, and suspected data breaches.
- ❖ Define escalation procedures and responsibilities for Security Operations Center (SOC) analysts.
- ❖ Conduct regular firewall rule reviews to remove obsolete or overly permissive rules.
- ❖ Perform routine vulnerability assessments and penetration testing to identify emerging security weaknesses.
- ❖ Maintain comprehensive network and system documentation to support incident response and disaster recovery activities.

These process improvements promote consistency, improve response times, and support regulatory compliance.

6.3 Priority 3: Enhance Monitoring and Detection

Continuous monitoring should be strengthened to improve threat detection and reduce false positives.

Recommended actions include:

- ❖ Tune **Wazuh SIEM** rules to prioritize high-risk events, including repeated authentication failures, privilege escalation attempts, unauthorized account creation, and network reconnaissance.
- ❖ Configure alert thresholds and behavioural baselines to reduce unnecessary alerts while improving the identification of abnormal activity.
- ❖ Expand log collection to include firewalls, servers, Active Directory, medical devices, VPN services, and critical applications.
- ❖ Integrate **File Integrity Monitoring (FIM)** to detect unauthorized modifications to sensitive system files.
- ❖ Enable real-time notification for high-severity alerts through email or other approved notification channels.
- ❖ Periodically review and update detection rules to address evolving cyber threats.

These measures improve visibility into the environment and support faster incident detection and response.

6.4 Priority 4: Long-Term Security Strategy

To maintain a strong cybersecurity posture, MediCore Health Systems should continue investing in long-term security improvements.

Recommended initiatives include:

- ❖ Adopt a **Zero Trust Security Architecture**, where every access request is continuously verified regardless of network location.
- ❖ Maintain a structured patch management program to ensure operating systems, applications, and medical devices receive timely security updates.
- ❖ Conduct regular cybersecurity awareness training to educate staff on phishing, password security, social engineering, and safe handling of Protected Health Information (PHI).
- ❖ Perform annual penetration testing and periodic security audits to validate the effectiveness of implemented controls.
- ❖ Implement continuous risk assessments to identify new threats associated with expanding healthcare technologies and connected medical devices.

- ❖ Review disaster recovery and business continuity plans regularly to ensure critical healthcare services remain available during cybersecurity incidents.

These long-term initiatives will strengthen organizational resilience, reduce cyber risk, and support compliance with healthcare security and privacy regulations such as **HIPAA**.

Table 6.1: Recommendation Priority Summary

Priority	Recommendation Area	Expected Benefit
High	SSH hardening, MFA, firewall restrictions, network segmentation	Prevent unauthorized access and lateral movement
High	Wazuh SIEM tuning and centralized log monitoring	Improve detection and incident response
Medium	Incident response playbooks, log retention, firewall reviews	Improve operational readiness and compliance
Medium	Vulnerability assessments and penetration testing	Identify and remediate security weaknesses proactively
Long-Term	Zero Trust Architecture, patch management, staff awareness training	Enhance organizational resilience and maintain long-term cybersecurity maturity

CHAPTER SEVEN

7.0 CONCLUSION

This project successfully designed and implemented a Security Operations Center (SOC) environment for MediCore Health Systems, a simulated healthcare organization that manages electronic health records (EHR), patient billing and insurance services, Internet of Medical Things (IoMT) devices, and an online patient portal. The network was designed using a segmented architecture consisting of LAN and DMZ networks protected by a pfSense firewall to improve security and limit unauthorized access between critical systems.

A Wazuh SIEM server was deployed to centrally monitor logs and security events from Windows workstations, Ubuntu servers, and IoMT devices. Kali Linux was used to simulate realistic cyberattacks, including network reconnaissance with Nmap and SSH brute-force attacks using Hydra. Network traffic generated during these attacks was captured and analyzed with Wireshark, while Wazuh successfully detected and logged security events, demonstrating the effectiveness of centralized monitoring and threat detection. Firewall rules were also implemented and reviewed to strengthen network security and control traffic between network segments.

The project provided practical experience in designing a secure healthcare network, deploying security monitoring tools, collecting and analyzing security logs, investigating attack activities, and applying security controls to reduce organizational risk. Overall, this SOC implementation demonstrated how multiple cybersecurity technologies can work together to improve the confidentiality, integrity, and availability of healthcare information systems while supporting effective incident detection and response.

CHAPTER EIGHT

8.0 References

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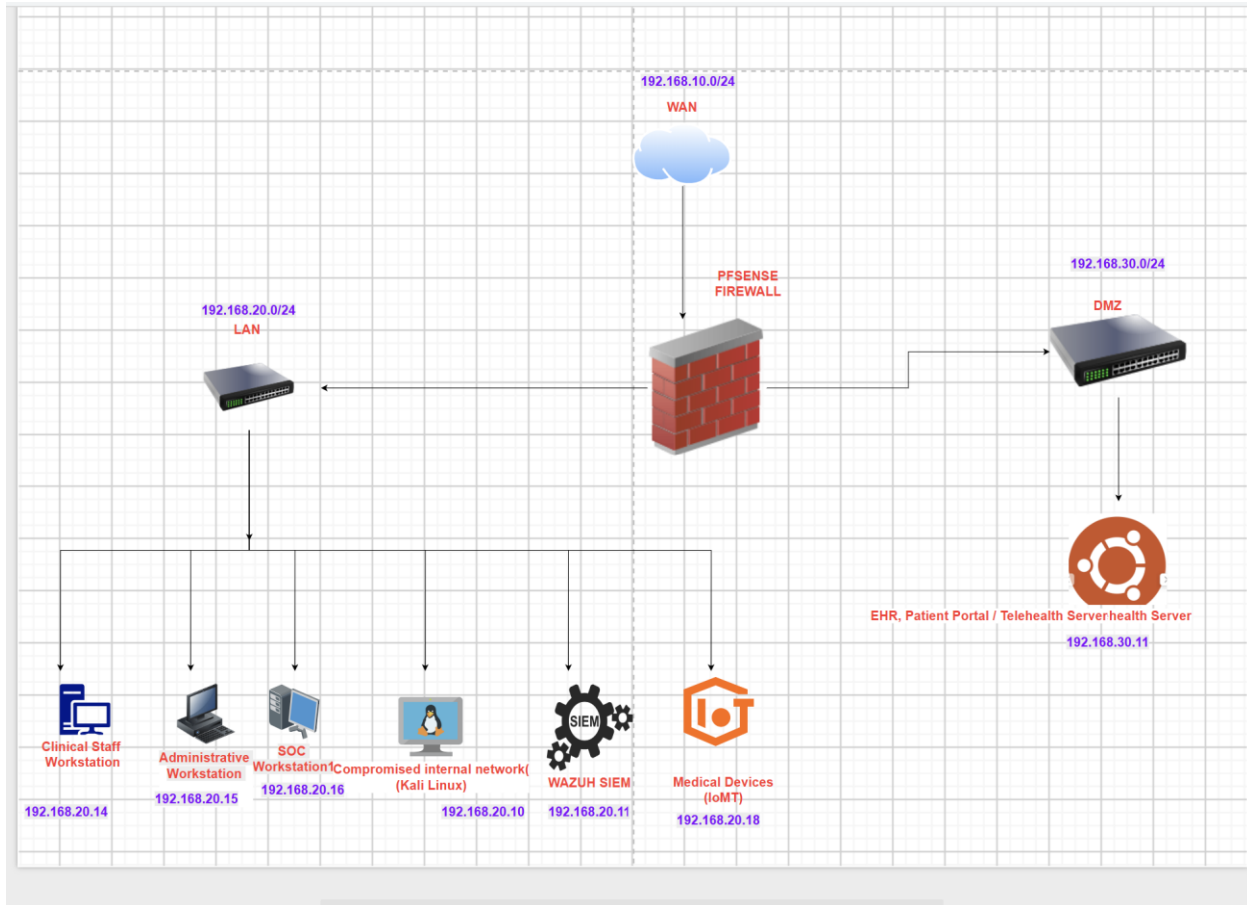
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CHAPTER NINE

9.0 Appendices

9.1 Appendix A — Network Design & Firewall Configuration

Enterprise Network Diagram created using Draw.io



Interface configuration on pfsense

The screenshot displays the pfSense web interface for configuring the LAN (em1) interface. The top navigation bar includes links for System, Interfaces, Firewall, Services, VPN, Status, Diagnostics, and Help. The main content area is titled 'Interfaces / LAN (em1)' and features a 'General Configuration' section. In this section, the 'Enable' checkbox is checked, and the 'Description' is set to 'LAN'. The 'IPv4 Configuration Type' is set to 'Static IPv4', and the 'IPv6 Configuration Type' is set to 'None'. The 'MAC Address' field is empty, with a note explaining its use for spoofing. The 'MTU' and 'MSS' fields are also empty, with explanatory text. The 'Speed and Duplex' is set to 'Default (no preference, typically autoselect)'. Below the general configuration is the 'Static IPv4 Configuration' section, where the 'IPv4 Address' is set to '192.168.20.1' and the 'IPv4 Upstream gateway' is set to 'None'. A green button labeled '+ Add a new gateway' is visible. The 'Reserved Networks' section is currently empty. The bottom screenshot shows the same interface but for the DMZ (em2) interface, with the 'Description' set to 'DMZ'.

Interfaces / LAN (em1)

General Configuration

Enable ☒ Enable interface

Description LAN
Enter a description (name) for the interface here.

IPv4 Configuration Type Static IPv4

IPv6 Configuration Type None

MAC Address xxxxxxxxxxxx
This field can be used to modify ("spoof") the MAC address of this interface.
Enter a MAC address in the following format: xxxxxxxxxxxx or leave blank.

MTU
If this field is blank, the adapter's default MTU will be used. This is typically 1500 bytes but can vary in some circumstances.

MSS
If a value is entered in this field, then MSS clamping for TCP connections to the value entered above minus 40 for IPv4 (TCP/IPv4 header size) and minus 60 for IPv6 (TCP/IPv6 header size) will be in effect.

Speed and Duplex Default (no preference, typically autoselect)
Explicitly set speed and duplex mode for this interface.
WARNING: MUST be set to autoselect (automatically negotiate speed) unless the port this interface connects to has its speed and duplex forced.

Static IPv4 Configuration

IPv4 Address 192.168.20.1 / 24

IPv4 Upstream gateway None + Add a new gateway

If this interface is an Internet connection, select an existing Gateway from the list or add a new one using the "Add" button.
On local area network interfaces the upstream gateway should be "none".
Selecting an upstream gateway causes the firewall to treat this interface as a [WAN type interface](#).
Gateways can be managed by [clicking here](#).

Reserved Networks

Interfaces / DMZ (em2)

General Configuration

Enable ☒ Enable interface

Description DMZ
Enter a description (name) for the interface here.

IPv4 Configuration Type Static IPv4

IPv6 Configuration Type None

MAC Address xxxxxxxxxxxx
This field can be used to modify ("spoof") the MAC address of this interface.
Enter a MAC address in the following format: xxxxxxxxxxxx or leave blank.

MTU
If this field is blank, the adapter's default MTU will be used. This is typically 1500 bytes but can vary in some circumstances.

MSS
If a value is entered in this field, then MSS clamping for TCP connections to the value entered above minus 40 for IPv4 (TCP/IPv4 header size) and minus 60 for IPv6 (TCP/IPv6 header size) will be in effect.

Speed and Duplex Default (no preference, typically autoselect)
Explicitly set speed and duplex mode for this interface.
WARNING: MUST be set to autoselect (automatically negotiate speed) unless the port this interface connects to has its speed and duplex forced.

pfSense Firewall Rules

pfSense
COMMUNITY EDITION

System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

7

Firewall / Rules / Edit

Firewall Rule

Action

Pass

Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

LAN

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

Any

Choose which IP protocol this rule should match.

Source

Source

☐ Invert match

LAN subnets

Source Address

/

Destination

Destination

☐ Invert match

Any

Destination Address

/

Extra Options

Log

☐ Log packets that are handled by this rule

Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the Status: System Logs: Settings page).

Description

Default allow LAN to any rule

A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset label and displayed in the firewall log.

Advanced Options

Display Advanced

Rule Information

Tracking ID

0100000101

Updated

7/28/26 17:41:03 by admin@192.168.20.10 (Local Database)

Save

pfSense
COMMUNITY EDITION

System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

7

Firewall / Rules / Edit

Firewall Rule

Action

Pass

Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

DMZ

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

Any

Choose which IP protocol this rule should match.

Source

Source

☐ Invert match

DMZ subnets

Source Address

/

Destination

Destination

☐ Invert match

Any

Destination Address

/

Extra Options

Log

☐ Log packets that are handled by this rule

Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the Status: System Logs: Settings page).

Description

A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset label and displayed in the firewall log.

Advanced Options

Display Advanced

pfSense Interface Assignments

pfSense COMMUNITY EDITION

System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

Interfaces / Interface Assignments

Interface Assignments Interface Groups Wireless VLANs QinQs PPPs GREs GIFs Bridges LAGGs

Interface	Network port
WAN	em0 (08:00:27:6d:56:0e)
LAN	em1 (08:00:27:81:19:bd) Delete
DMZ	em2 (08:00:27:6c:74:04) Delete

Save

Interfaces that are configured as members of a lagg(4) interface will not be shown.
Wireless interfaces must be created on the Wireless tab before they can be assigned.

DHCP server configuration for LAN on pfsense.

pfSense COMMUNITY EDITION

System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

Services / DHCP Server / LAN

ISC DHCP has reached end-of-life and will be removed in a future version of pfSense. Visit [System > Advanced > Networking](#) to switch DHCP backend.

LAN DMZ

General Settings

DHCP Backend	ISC DHCP
Enable	☒ Enable DHCP server on LAN interface
BOOTP	☐ Ignore BOOTP queries
Deny Unknown Clients	Allow all clients When set to Allow all clients, any DHCP client will get an IP address within this scope/range on this interface. If set to Allow known clients from any interface, any DHCP client with a MAC address listed in a static mapping on any scope(s)/interface(s) will get an IP address. If set to Allow known clients from only this interface, only MAC addresses listed in static mappings on this interface will get an IP address within this scope/range.
Ignore Denied Clients	☐ Ignore denied clients rather than reject This option is not compatible with failover and cannot be enabled when a Failover Peer IP address is configured.
Ignore Client Identifiers	☐ Do not record a unique identifier (UID) in client lease data if present in the client DHCP request This option may be useful when a client can dual boot using different client identifiers but the same hardware (MAC) address. Note that the resulting server behavior violates the official DHCP specification.

Primary Address Pool

Subnet	192.168.20.0/24
Subnet Range	192.168.20.1 - 192.168.20.254
Address Pool Range	192.168.20.10 From 192.168.20.100 To The specified range for this pool must not be within the range configured on any other address pool for this interface.
Additional Pools	+ Add Address Pool If additional pools of addresses are needed inside of this subnet outside the above range, they may be specified here.

DHCP server configuration for DMZ on pfsense.

The screenshot shows the pfSense web interface for the DHCP Server configuration on the DMZ interface. The top navigation bar includes links for System, Interfaces, Firewall, Services, VPN, Status, Diagnostics, and Help. A notification banner at the top states: "ISC DHCP has reached end-of-life and will be removed in a future version of pfSense. Visit [System > Advanced > Networking](#) to switch DHCP backend."

The configuration is divided into three main sections:

- General Settings:**
 - DHCP Backend:** ISC DHCP
 - Enable:** ☒ Enable DHCP server on DMZ interface
 - BOOTP:** ☐ Ignore BOOTP queries
 - Deny Unknown Clients:** ☐ Allow all clients. When set to **Allow all clients**, any DHCP client will get an IP address within this scope/range on this interface. If set to **Allow known clients from any interface**, any DHCP client with a MAC address listed in a static mapping on **any** scope(s)/interface(s) will get an IP address. If set to **Allow known clients from only this interface**, only MAC addresses listed in static mappings on this interface will get an IP address within this scope/range.
 - Ignore Denied Clients:** ☐ Ignore denied clients rather than reject. This option is not compatible with failover and cannot be enabled when a Failover Peer IP address is configured.
 - Ignore Client Identifiers:** ☐ Do not record a unique identifier (UID) in client lease data if present in the client DHCP request. This option may be useful when a client can dual boot using different client identifiers but the same hardware (MAC) address. Note that the resulting server behavior violates the official DHCP specification.
- Primary Address Pool:**
 - Subnet:** 192.168.30.0/24
 - Subnet Range:** 192.168.30.1 - 192.168.30.254
 - Address Pool Range:** From 192.168.30.10 To 192.168.30.100. The specified range for this pool must not be within the range configured on any other address pool for this interface.
 - Additional Pools:** [+ Add Address Pool](#). If additional pools of addresses are needed inside of this subnet outside the above range, they may be specified here.
- Server Options:**
 - WINS Servers:** WINS Server 1, WINS Server 2

pfSense start-up page showing all the interface configured.

```
Starting CRON... done.
pfSense 2.8.1-RELEASE amd64 20260513-1943
Bootup complete

FreeBSD/amd64 (MedCore.home.arpa) (ttyv0)

VirtualBox Virtual Machine - Netgate Device ID: 78b16881aafda2a786a8

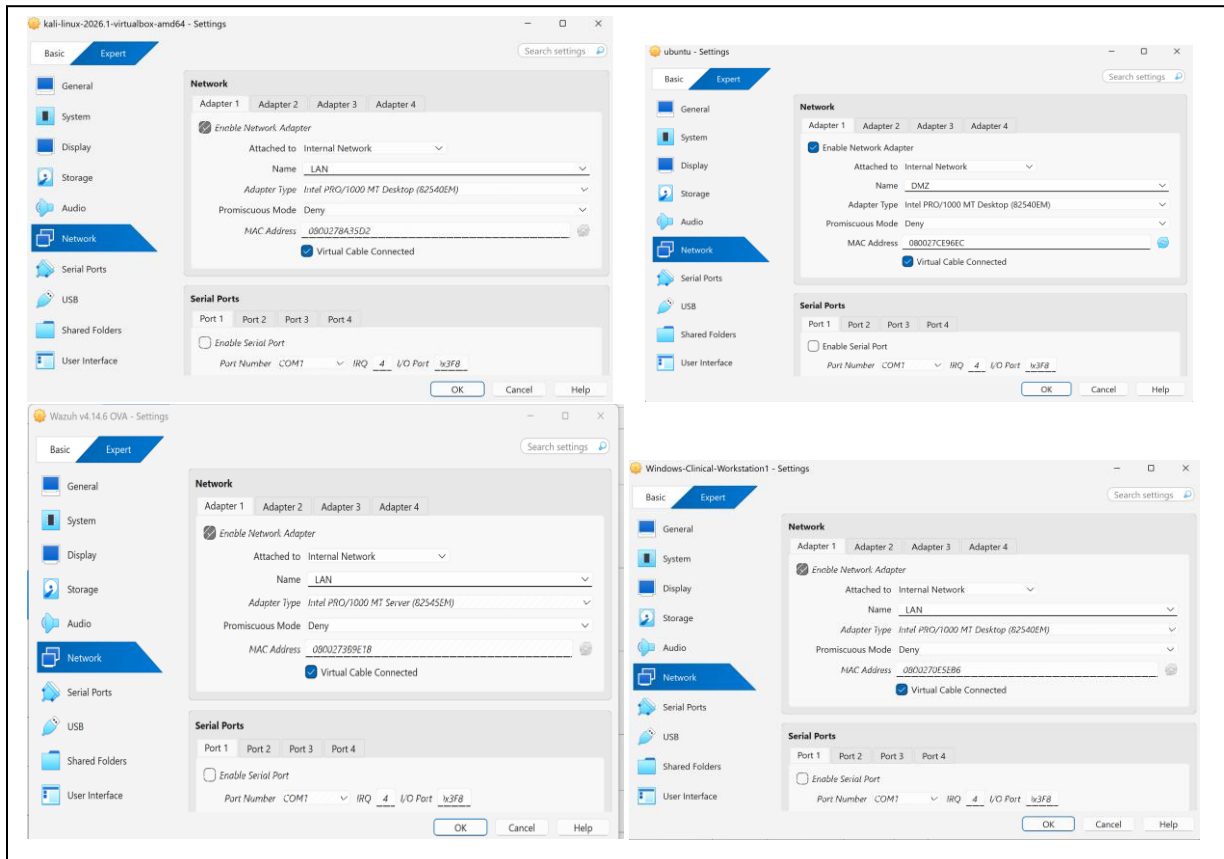
*** Welcome to pfSense 2.8.1-RELEASE (amd64) on MedCore ***

WAN (wan) -> em0 -> v4/DHCP4: 192.168.10.3/24
LAN (lan) -> em1 -> v4: 192.168.20.1/24
DMZ (opt1) -> em2 -> v4: 192.168.30.1/24

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address      11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults         13) Update from console
5) Reboot system                    14) Enable Secure Shell (sshd)
6) Halt system                      15) Restore recent configuration
7) Ping host                        16) Restart PHP-FPM
8) Shell

Enter an option: 
```

Asset assignment to their respective security zones



9.2 Appendix B — Attack Simulation & Traffic Analysis

Kali Linux terminal showing Nmap scan results

```
(kali㉿kali)-[~]
$ nmap -sV 192.168.20.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-03 23:18 -0400
Nmap scan report for MediCore.home.arpa (192.168.20.1)
Host is up (0.00036s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
53/tcp    open  domain  Unbound
80/tcp    open  http    nginx
443/tcp   open  ssl/http nginx
MAC Address: 08:00:27:81:19:BD (Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 16.80 seconds

(kali㉿kali)-[~]
$ nmap -sV 192.168.20.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-03 23:19 -0400
Nmap scan report for MediCore.home.arpa (192.168.20.1)
Host is up (0.00045s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
53/tcp    open  domain  Unbound
80/tcp    open  http    nginx
443/tcp   open  ssl/http nginx
MAC Address: 08:00:27:81:19:BD (Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 17.39 seconds

(kali㉿kali)-[~]
$ nmap 192.168.30.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-03 23:19 -0400
Nmap scan report for 192.168.30.1
Host is up (0.00041s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  https

Nmap scan report for 192.168.30.11
Host is up (0.0012s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh

Nmap done: 256 IP addresses (2 hosts up) scanned in 9.43 seconds

(kali㉿kali)-[~]
$ nmap -sU 192.168.20.1 -p 161 --script=snmp.interfaces 192.168.20.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-03 23:20 -0400
Nmap scan report for MediCore.home.arpa (192.168.20.1)
Host is up (0.00072s latency).
```

```

$ nmap -sU 192.168.20.1 -p 161 --script=snmp.interfaces 192.168.20.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-03 23:20 -0400
Nmap scan report for MediCore.home.arpa (192.168.20.1)
Host is up (0.00072s latency).

PORT      STATE SERVICE
161/udp   open  snmp
| snmp.interfaces:
|   em0
|     IP address: 192.168.10.3 Netmask: 255.255.255.0
|     MAC address: 08:00:27:6d:56:0e (Oracle VirtualBox virtual NIC)
|     Type: ethernetCsmacd Speed: 1 Gbps
|     Status: up
|     Traffic stats: 818.99 Kb sent, 4.89 Mb received
|   em1
|     IP address: 192.168.20.1 Netmask: 255.255.255.0
|     MAC address: 08:00:27:81:19:bd (Oracle VirtualBox virtual NIC)
|     Type: ethernetCsmacd Speed: 1 Gbps
|     Status: up
|     Traffic stats: 3.95 Mb sent, 1.51 Mb received
|   em2
|     IP address: 192.168.30.1 Netmask: 255.255.255.0
|     MAC address: 08:00:27:6c:74:04 (Oracle VirtualBox virtual NIC)
|     Type: ethernetCsmacd Speed: 1 Gbps
|     Status: up
|     Traffic stats: 1.56 Mb sent, 1.06 Mb received
|   enc0
|     Type: wwanPP2 Speed: 0 Kbps
|     Status: down
|     Traffic stats: 0.00 Kb sent, 0.00 Kb received
|   lo0
|     IP address: 127.0.0.1 Netmask: 255.0.0.0
|     Type: softwareLoopback Speed: 0 Kbps
|     Status: up
|     Traffic stats: 3.93 Kb sent, 3.93 Kb received
|   pflog0
|     Type: ifPwType Speed: 0 Kbps
|     Status: down
|     Traffic stats: 20.32 Kb sent, 0.00 Kb received
|   pfsync0
|     Type: ilan Speed: 0 Kbps
|     Status: down
|     Traffic stats: 0.00 Kb sent, 0.00 Kb received
|_ MAC Address: 08:00:27:81:19:BD (Oracle VirtualBox virtual NIC)

Nmap scan report for MediCore.home.arpa (192.168.20.1)
Host is up (0.00028s latency).

PORT      STATE SERVICE
161/udp   open  snmp
| snmp.interfaces:
|   em0
|     IP address: 192.168.10.3 Netmask: 255.255.255.0
|     MAC address: 08:00:27:6d:56:0e (Oracle VirtualBox virtual NIC)
|     Type: ethernetCsmacd Speed: 1 Gbps
|     Status: up

```

```
File Machine View Input Devices Help

(kali@kali)-[~]
└─$ nmap -sn 192.168.20.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-05 05:31 -0400
Nmap scan report for MediCore.home.arpa (192.168.20.1)
Host is up (0.00038s latency).
MAC Address: 08:00:27:81:19:BD (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.20.11
Host is up (0.00032s latency).
MAC Address: 08:00:27:2B:29:CB (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.20.10
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 2.48 seconds

(kali@kali)-[~]
└─$ nmap -sn 192.168.30.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-05 05:31 -0400
Nmap scan report for 192.168.30.1
Host is up (0.00055s latency).
Nmap scan report for 192.168.30.11
Host is up (0.00058s latency).
Nmap done: 256 IP addresses (2 hosts up) scanned in 4.54 seconds

(kali@kali)-[~]
└─$ nmap -sS 192.168.30.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-05 05:32 -0400
Nmap scan report for 192.168.30.1
Host is up (0.00038s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  https

Nmap done: 1 IP address (1 host up) scanned in 5.36 seconds

(kali@kali)-[~]
└─$ nmap -sV 192.168.30.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-05 05:32 -0400
Nmap scan report for 192.168.30.1
Host is up (0.00034s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
53/tcp    open  domain Unbound
80/tcp    open  http  nginx
443/tcp   open  ssl/http nginx

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 18.08 seconds

(kali@kali)-[~]
└─$ nmap -sU 192.168.30.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-05 05:33 -0400
Nmap scan report for 192.168.30.1
Host is up (0.00041s latency).
Not shown: 997 open|filtered udp ports (no-response)
PORT      STATE SERVICE
53/udp    open  domain
123/udp   open  ntp
161/udp   open  snmp

Nmap done: 1 IP address (1 host up) scanned in 8.20 seconds

(kali@kali)-[~]
```

```

(kali@kali)-[~]
$ nmap 192.168.30.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-05 06:18 -0400
Nmap scan report for 192.168.30.1
Host is up (0.00036s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  https

Nmap scan report for 192.168.30.11
Host is up (0.00080s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh

Nmap done: 256 IP addresses (2 hosts up) scanned in 9.94 seconds

(kali@kali)-[~]

```

Kali Linux terminal showing Hydra SSH brute-force execution

```

[DATA] attacking ssh://192.168.30.10:22/
[ERROR] could not connect to ssh://192.168.30.10:22 - Timeout connecting to 192.168.30.10

(kali@kali)-[~]
$ ping 192.168.30.10
PING 192.168.30.10 (192.168.30.10) 56(84) bytes of data.
^C
--- 192.168.30.10 ping statistics ---
169 packets transmitted, 0 received, 100% packet loss, time 172437ms

(kali@kali)-[~]
$ hydra -L users.txt -P password.txt -t 4 -V ssh://192.168.30.11
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-08-03 23:11:20
[DATA] max 4 tasks per 1 server, overall 4 tasks, 24 login tries (l:6/p:4), -6 tries per task
[DATA] attacking ssh://192.168.30.11:22/
[ATTEMPT] target 192.168.30.11 - login "admin" - pass "password" - 1 of 24 [child 0] (0/0)
[ATTEMPT] target 192.168.30.11 - login "admin" - pass "ubuntu" - 2 of 24 [child 1] (0/0)
[ATTEMPT] target 192.168.30.11 - login "admin" - pass "123456789" - 3 of 24 [child 2] (0/0)
[ATTEMPT] target 192.168.30.11 - login "admin" - pass "10Alytics" - 4 of 24 [child 3] (0/0)
[22][ssh] host: 192.168.30.11 login: admin password: password
[ATTEMPT] target 192.168.30.11 - login "ubuntu" - pass "password" - 5 of 24 [child 0] (0/0)
[ATTEMPT] target 192.168.30.11 - login "ubuntu" - pass "ubuntu" - 6 of 24 [child 2] (0/0)
[ATTEMPT] target 192.168.30.11 - login "ubuntu" - pass "123456789" - 7 of 24 [child 1] (0/0)
[ATTEMPT] target 192.168.30.11 - login "ubuntu" - pass "10Alytics" - 8 of 24 [child 3] (0/0)
[22][ssh] host: 192.168.30.11 login: ubuntu password: ubuntu
[ATTEMPT] target 192.168.30.11 - login "medicore" - pass "password" - 9 of 24 [child 2] (0/0)
[ATTEMPT] target 192.168.30.11 - login "medicore" - pass "ubuntu" - 10 of 24 [child 0] (0/0)
[22][ssh] host: 192.168.30.11 login: medicore password: password
[ATTEMPT] target 192.168.30.11 - login "johnson" - pass "password" - 13 of 24 [child 2] (0/0)
[ATTEMPT] target 192.168.30.11 - login "johnson" - pass "ubuntu" - 14 of 24 [child 1] (0/0)
[ATTEMPT] target 192.168.30.11 - login "johnson" - pass "123456789" - 15 of 24 [child 3] (0/0)
[ATTEMPT] target 192.168.30.11 - login "johnson" - pass "10Alytics" - 16 of 24 [child 2] (0/0)
[ATTEMPT] target 192.168.30.11 - login "10Alytics" - pass "password" - 17 of 24 [child 0] (0/0)
[ATTEMPT] target 192.168.30.11 - login "10Alytics" - pass "ubuntu" - 18 of 24 [child 1] (0/0)
[ATTEMPT] target 192.168.30.11 - login "10Alytics" - pass "123456789" - 19 of 24 [child 3] (0/0)
[ATTEMPT] target 192.168.30.11 - login "10Alytics" - pass "10Alytics" - 20 of 24 [child 2] (0/0)
[ATTEMPT] target 192.168.30.11 - login "" - pass "password" - 21 of 24 [child 0] (0/0)
[ATTEMPT] target 192.168.30.11 - login "" - pass "ubuntu" - 22 of 24 [child 1] (0/0)
[ATTEMPT] target 192.168.30.11 - login "" - pass "123456789" - 23 of 24 [child 3] (0/0)
[ATTEMPT] target 192.168.30.11 - login "" - pass "10Alytics" - 24 of 24 [child 2] (0/0)
1 of 1 target successfully completed, 3 valid passwords found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-08-03 23:11:32

(kali@kali)-[~]

```

The image shows a Wireshark packet capture of ICMP Echo (ping) traffic. The packet list on the left shows 20 packets. The packet details pane on the right shows the structure of the selected packet (No. 10). The packet bytes pane at the bottom shows the raw data of the selected packet.

No.	Time	Source	Destination	Protocol	Len	Info
10.0.000.0	192.168.20.10	192.168.10.10	ICMP	42	Echo (ping) request	id=0xa3e7, seq=0/0, ttl=39 (no response found!)
1.1.1231.	192.168.20.10	192.168.10.10	ICMP	42	Echo (ping) request	id=0x1b74, seq=0/0, ttl=55 (no response found!)
7.2.2341.	192.168.20.10	192.168.10.10	TCP	58	34674 → 443 [SYN] Seq=0	Win=1024 Len=0 MSS=1460
7.2.3388.	192.168.20.10	192.168.10.10	TCP	58	34674 → 443 [SYN] Seq=0	Win=1024 Len=0 MSS=1460
1.3.2655.	192.168.20.10	192.168.10.10	TCP	54	34674 → 80 [ACK] Seq=1	Ack=1 Win=1024 Len=0
1.3.3664.	192.168.20.10	192.168.10.10	TCP	54	34674 → 80 [ACK] Seq=1	Ack=1 Win=1024 Len=0
1.3.6769.	192.168.20.10	192.168.10.10	ICMP	54	Timestamp request	id=0xf6ce, seq=0/0, ttl=48
1.3.7777.	192.168.20.10	192.168.10.10	ICMP	54	Timestamp request	id=0x8144, seq=0/0, ttl=58
1.868.89.	192.168.20.10	192.168.10.10	ICMP	42	Echo (ping) request	id=0x812f, seq=0/0, ttl=49 (no response found!)
1.870.86.	192.168.20.10	192.168.10.10	ICMP	42	Echo (ping) request	id=0x3237, seq=0/0, ttl=38 (no response found!)
1.870.53.	192.168.20.10	192.168.10.10	TCP	58	39538 → 443 [SYN] Seq=0	Win=1024 Len=0 MSS=1460
1.870.64.	192.168.20.10	192.168.10.10	TCP	58	39540 → 443 [SYN] Seq=0	Win=1024 Len=0 MSS=1460
1.870.83.	192.168.20.10	192.168.10.10	TCP	54	39538 → 80 [ACK] Seq=1	Ack=1 Win=1024 Len=0
1.870.96.	192.168.20.10	192.168.10.10	TCP	54	39540 → 80 [ACK] Seq=1	Ack=1 Win=1024 Len=0
1.871.88.	192.168.20.10	192.168.10.10	ICMP	54	Timestamp request	id=0x19bf, seq=0/0, ttl=48
1.871.22.	192.168.20.10	192.168.10.10	ICMP	54	Timestamp request	id=0x5842, seq=0/0, ttl=58
1.994.54.	192.168.20.10	192.168.10.10	ICMP	42	Echo (ping) request	id=0xb112, seq=0/0, ttl=49 (no response found!)
1.995.75.	192.168.20.10	192.168.10.10	ICMP	42	Echo (ping) request	id=0x46ee, seq=0/0, ttl=39 (no response found!)
1.995.89.	192.168.20.10	192.168.10.10	TCP	58	46374 → 443 [SYN] Seq=0	Win=1024 Len=0 MSS=1460
1.996.89.	192.168.20.10	192.168.10.10	TCP	58	46376 → 443 [SYN] Seq=0	Win=1024 Len=0 MSS=1460
1.996.19.	192.168.20.10	192.168.10.10	TCP	54	46374 → 80 [ACK] Seq=1	Ack=1 Win=1024 Len=0
1.996.29.	192.168.20.10	192.168.10.10	TCP	54	46376 → 80 [ACK] Seq=1	Ack=1 Win=1024 Len=0
1.996.29.	192.168.20.10	192.168.10.10	ICMP	54	Timestamp request	id=0x442c, seq=0/0, ttl=40
1.996.39.	192.168.20.10	192.168.10.10	ICMP	54	Timestamp request	id=0xb3c2, seq=0/0, ttl=54

Frame 753: 58 bytes on wire (464 bits), 58 bytes captured (464 bits) on interface eth0, id 0
 Ethernet II, Src: PCSystemtec_8a:35:d2 (08:00:27:8a:35:d2), Dst: PCSystemtec_81:19:bd (08:00:27:81:19:bd)
 Internet Protocol Version 4, Src: 192.168.20.10, Dst: 192.168.10.10
 Transmission Control Protocol, Src Port: 34676, Dst Port: 443, Seq: 0, Len: 0

0000 08 00 27 81 19 bd 08 00 27 8a 35 d2 08 00 45 00 5 ..E
 0010 00 2c 54 14 00 00 38 06 7b 53 c8 a8 14 0a c8 a8 T ..8 (\$
 0020 1e 8a 87 74 01 bb fb c2 33 c1 00 00 00 00 00 02 3
 0030 04 00 28 9e 00 00 02 04 05 04 (.....
 0040

kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Capturing from eth0
 192.168.30.1 & tcp port == 22

No.	Time	Source	Destination	Protocol	Length	Info
9130	1488.6345599	192.168.20.10	192.168.20.10	SSMv2	109	Server: Protocol (SSH-2.0-openssh.9.6p1 Ubuntu-Subunit13.18)
9130	1488.6345999	192.168.20.10	192.168.20.11	TCP	60	44816 -> 22 [ACK] Seq=24 Ack=44 Min=64512 Len=0 TSval=2316274149 TSecr=2581143221
9140	1488.6346927	192.168.20.10	192.168.20.11	SSMv2	1042	Client: Key Exchange Init
9141	1488.6350964	192.168.20.11	192.168.20.10	SSMv2	1186	Server: Key Exchange Init
9142	1488.6374731	192.168.20.10	192.168.20.11	SSMv2	114	Client: Elliptic Curve Diffie-Hellman Key Exchange Init
9151	1488.6373240	192.168.20.10	192.168.20.11	SSMv2	1186	Server: Key Exchange Reply, New Keys, Encrypted packet (len=284)
9154	1488.6374112	192.168.20.10	192.168.20.11	SSMv2	114	Client: Elliptic Curve Diffie-Hellman Key Exchange Init
9145	1488.6418984	192.168.20.11	192.168.20.10	SSMv2	558	Server: Elliptic Curve Diffie-Hellman Key Exchange Reply, New Keys, Encrypted packet (len=284)
9146	1488.6418987	192.168.20.10	192.168.20.11	SSMv2	82	Client: New Keys
9147	1488.6427017	192.168.20.11	192.168.20.10	SSMv2	558	Server: Elliptic Curve Diffie-Hellman Key Exchange Reply, New Keys, Encrypted packet (len=284)
9148	1488.6438919	192.168.20.10	192.168.20.11	SSMv2	82	Client: New Keys
9149	1488.6836045	192.168.20.10	192.168.20.11	TCP	66	22 -> 44816 [ACK] Seq=1656 Ack=1064 Win=64512 Len=0 TSval=2581143270 TSecr=2316274157
9150	1488.6836048	192.168.20.11	192.168.20.10	TCP	66	22 -> 44822 [ACK] Seq=1656 Ack=1064 Win=64512 Len=0 TSval=2429288156 TSecr=2987966425
9151	1488.6836240	192.168.20.10	192.168.20.11	SSMv2	110	Client: Encrypted packet (len=64)
9152	1488.6836368	192.168.20.10	192.168.20.11	SSMv2	110	Client: Encrypted packet (len=64)
9153	1488.6836772	192.168.20.10	192.168.20.11	TCP	66	22 -> 44816 [ACK] Seq=1656 Ack=1108 Win=64512 Len=0 TSval=2581143270 TSecr=2316274158
9154	1488.6836773	192.168.20.10	192.168.20.11	TCP	66	22 -> 44822 [ACK] Seq=1656 Ack=1108 Win=64512 Len=0 TSval=2429288156 TSecr=2987966425
9155	1488.6841884	192.168.20.11	192.168.20.10	SSMv2	110	Server: Encrypted packet (len=64)
9156	1488.6842197	192.168.20.10	192.168.20.11	SSMv2	110	Server: Encrypted packet (len=64)
9157	1488.6842628	192.168.20.10	192.168.20.11	TCP	126	Client: Encrypted packet (len=60)
9158	1488.6843176	192.168.20.10	192.168.20.11	SSMv2	126	Client: Encrypted packet (len=60)
9159	1488.6888485	192.168.20.10	192.168.20.11	SSMv2	118	Server: Encrypted packet (len=52)
9160	1488.6881505	192.168.20.10	192.168.20.11	SSMv2	150	Client: Encrypted packet (len=84)
9161	1488.6901284	192.168.20.10	192.168.20.11	SSMv2	118	Server: Encrypted packet (len=84)
9162	1488.6901996	192.168.20.10	192.168.20.11	TCP	142	Client: Encrypted packet (len=76)
9163	1488.7294805	192.168.20.10	192.168.20.11	SSMv2	118	Server: Encrypted packet (len=52)
9164	1488.7313435	192.168.20.11	192.168.20.10	TCP	66	22 -> 44816 [ACK] Seq=1752 Ack=1244 Win=64512 Len=0 TSval=2429288156 TSecr=2987966425
9165	1488.7315536	192.168.20.11	192.168.20.10	SSMv2	118	Server: Encrypted packet (len=52)
9166	1488.7321388	192.168.20.10	192.168.20.11	TCP	118	Client: Encrypted packet (len=52)
9167	1488.7321388	192.168.20.10	192.168.20.11	TCP	60	44816 -> 22 [FIN, ACK] Seq=1313231313 Win=65536 Len=0 TSval=18202272046 TSecr=1878976332
9168	1488.7321388	192.168.20.10	192.168.20.11	TCP	74	44816 -> 22 [RST, Seq=1878976332, Win=0, MSS=65536, SACK_PERM, TSval=18202272046 TSecr=0, WS=256
9169	1488.7321388	192.168.20.11	192.168.20.10	TCP	66	22 -> 44816 [ACK] Seq=1804 Ack=1312 Win=64512 Len=0 TSval=17199976333 TSecr=18202272046
9170	1488.7321388	192.168.20.11	192.168.20.10	TCP	74	

Relevant packet details highlighting source IP address, destination IP address, protocol, ports, and timestamps.

No.	Time	Source	Destination	Proto	Len	Info
10	0.0000	192.168.20.192	192.168.30.192	ICMP	42	Echo (ping) request id=0xa367, seq=0/0, ttl=39 (no response found)
1	1.1231	192.168.20.192	192.168.30.192	ICMP	42	Echo (ping) request id=0xb74, seq=0/0, ttl=55 (no response found)
7	2.2341	192.168.20.192	192.168.30.192	TCP	58	34674 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2	2.3101	192.168.20.192	192.168.30.192	TCP	58	34674 → 443 [ACK] Seq=0 Win=1024 Len=0 MSS=1460
1	3.2655	192.168.20.192	192.168.30.192	TCP	54	34674 → 80 [ACK] Seq=1 Ack=1 Win=1024 Len=0
1	3.3664	192.168.20.192	192.168.30.192	TCP	54	34676 → 80 [ACK] Seq=1 Ack=1 Win=1024 Len=0
1	3.6769	192.168.20.192	192.168.30.192	ICMP	54	Timestamp request id=0xf6e, seq=0/0, ttl=48
1	3.7777	192.168.20.192	192.168.30.192	ICMP	54	Timestamp request id=0x8144, seq=0/0, ttl=58
1	868.89	192.168.20.192	192.168.30.192	ICMP	42	Echo (ping) request id=0x812f, seq=0/0, ttl=49 (no response found)
1	870.86	192.168.20.192	192.168.30.192	ICMP	42	Echo (ping) request id=0x3237, seq=0/0, ttl=38 (no response found)
1	870.53	192.168.20.192	192.168.30.192	TCP	58	39538 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1	870.64	192.168.20.192	192.168.30.192	TCP	58	39540 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1	870.63	192.168.20.192	192.168.30.192	TCP	54	39538 → 80 [ACK] Seq=1 Ack=1 Win=1024 Len=0
1	870.96	192.168.20.192	192.168.30.192	TCP	54	39540 → 80 [ACK] Seq=1 Ack=1 Win=1024 Len=0
1	871.06	192.168.20.192	192.168.30.192	ICMP	54	Timestamp request id=0x19bf, seq=0/0, ttl=48
1	871.22	192.168.20.192	192.168.30.192	ICMP	54	Timestamp request id=0x5842, seq=0/0, ttl=58
1	994.54	192.168.20.192	192.168.30.192	ICMP	42	Echo (ping) request id=0xb112, seq=0/0, ttl=49 (no response found)
1	995.75	192.168.20.192	192.168.30.192	ICMP	42	Echo (ping) request id=0x46ee, seq=0/0, ttl=39 (no response found)
1	996.99	192.168.20.192	192.168.30.192	TCP	58	46374 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1	996.99	192.168.20.192	192.168.30.192	TCP	58	46376 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1	996.19	192.168.20.192	192.168.30.192	TCP	54	46374 → 80 [ACK] Seq=1 Ack=1 Win=1024 Len=0
1	996.29	192.168.20.192	192.168.30.192	TCP	54	46376 → 80 [ACK] Seq=1 Ack=1 Win=1024 Len=0
1	996.29	192.168.20.192	192.168.30.192	ICMP	54	Timestamp request id=0x442c, seq=0/0, ttl=40
1	996.39	192.168.20.192	192.168.30.192	ICMP	54	Timestamp request id=0xbc32, seq=0/0, ttl=54

Frame 753: Packet, 58 bytes on wire (464 bits), 58 bytes captured (464 bits) on interface eth0, id 0

Ethernet II, Src: PCSSystemtec_8a:35:d2 (08:00:27:8a:35:d2), Dst: PCSSystemtec_81:19:bd (08:00:27:81:19:bd)

Internet Protocol Version 4, Src: 192.168.20.192, Dst: 192.168.30.192

Transmission Control Protocol, Src Port: 34676, Dst Port: 443, Seq: 0, Len: 0

0000 08 00 27 81 19 bd 08 00 27 8a 35 d2 08 00 45 00

0010 00 2c 54 14 00 00 3b 05 7b 53 c0 a5 14 8a c0 a5

0020 1e 8a 87 74 01 0b 7b c2 33 c1 00 00 00 00 68 02

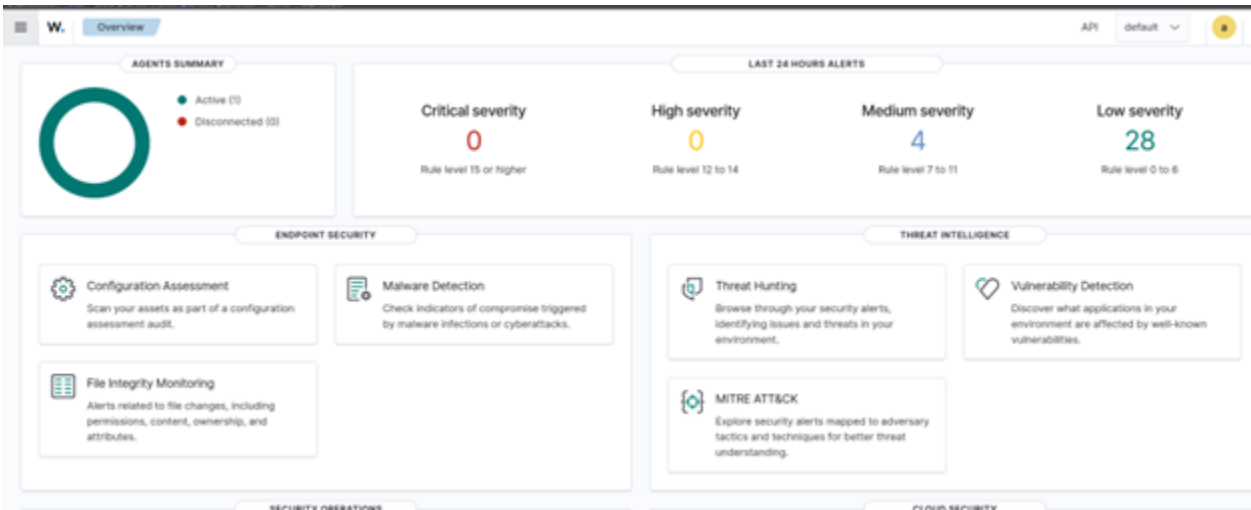
0030 04 00 28 0e 00 00 02 04 05 b4

9.30 Appendix C — Wazuh SIEM Detection & Response

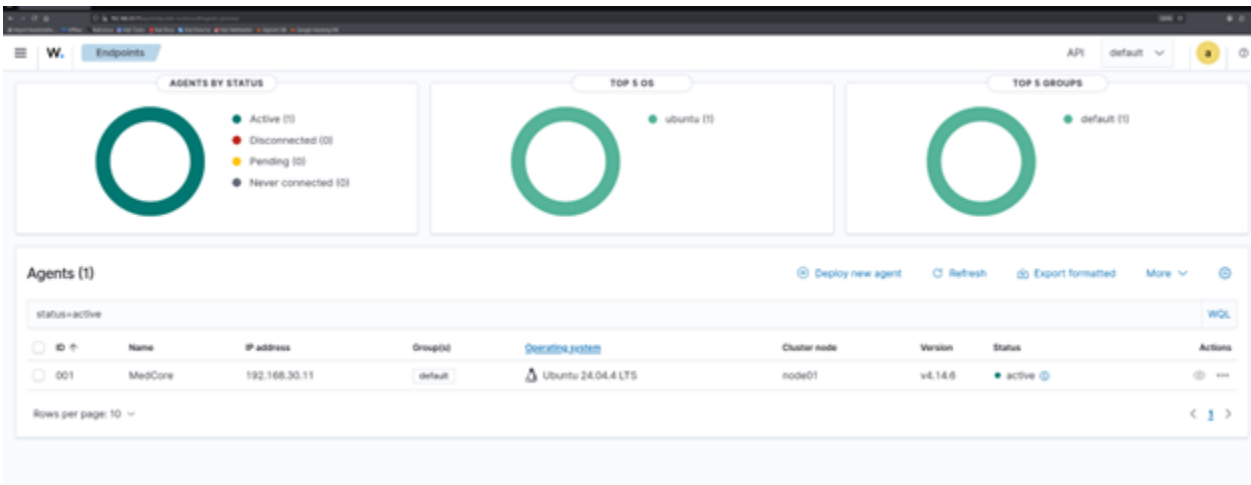
Appendix C

Include the following supporting evidence:

- Wazuh Dashboard overview



- Registered agent status



- Authentication failure alerts

Table JSON

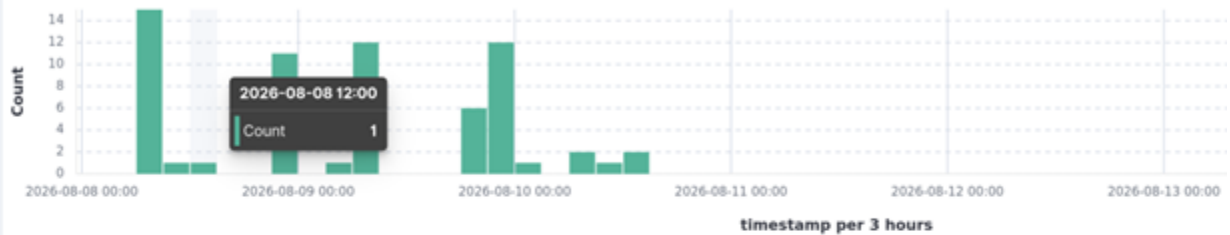
f _index	wazuh-alerts-4.x-2026.08.10
f agent.id	001
f agent.ip	192.168.20.11
f agent.name	MedCore
f data.euid	0
f data.srcip	192.168.20.10
f data.tty	ssh
f data.uid	0
f _decoder.name	nam

Table JSON

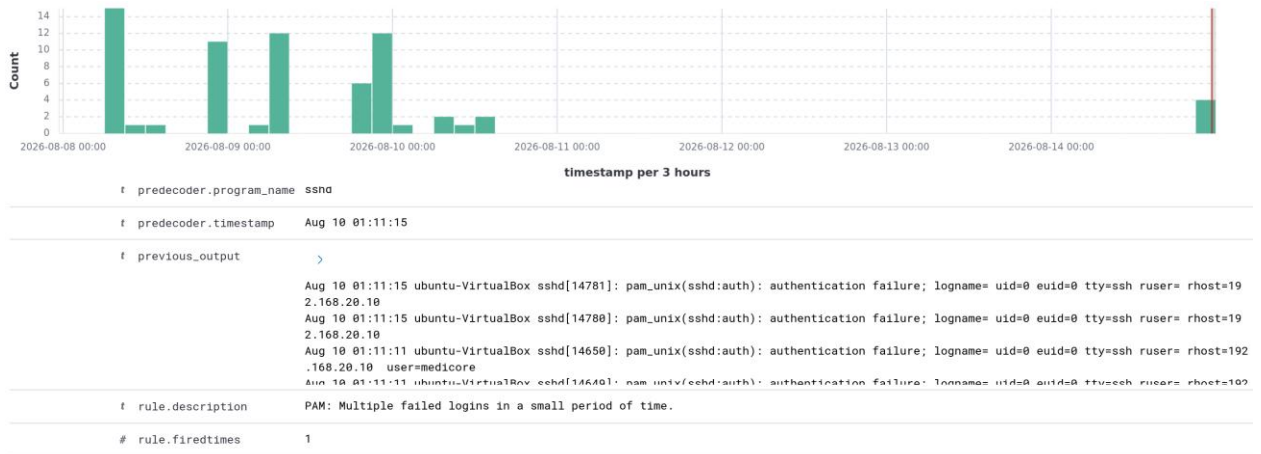
f _index	wazuh-alerts-4.x-2026.08.10
f agent.id	001
f agent.ip	192.168.20.11
f agent.name	MedCore
f data.euid	0
f data.srcip	192.168.20.10
f data.tty	ssh
f data.uid	0
f _decoder.name	nam

Aug 7, 2026 @ 23:24:47.762 - Aug 14, 2026 @ 23:24:47.763 per

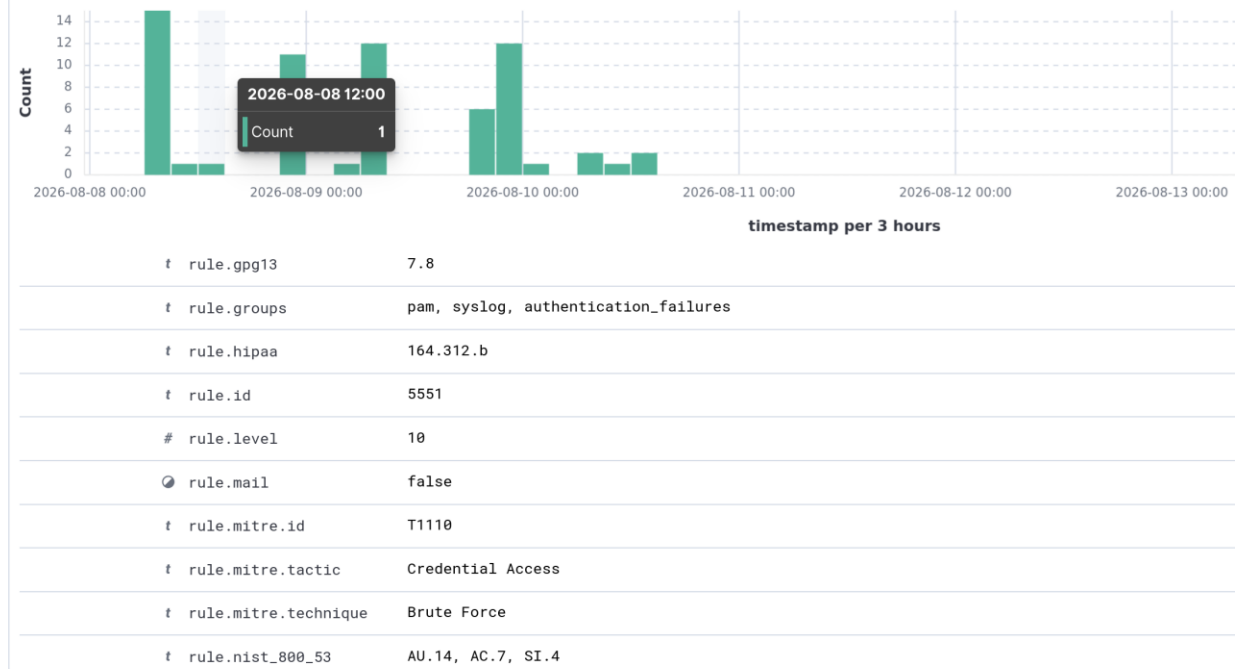
Auto

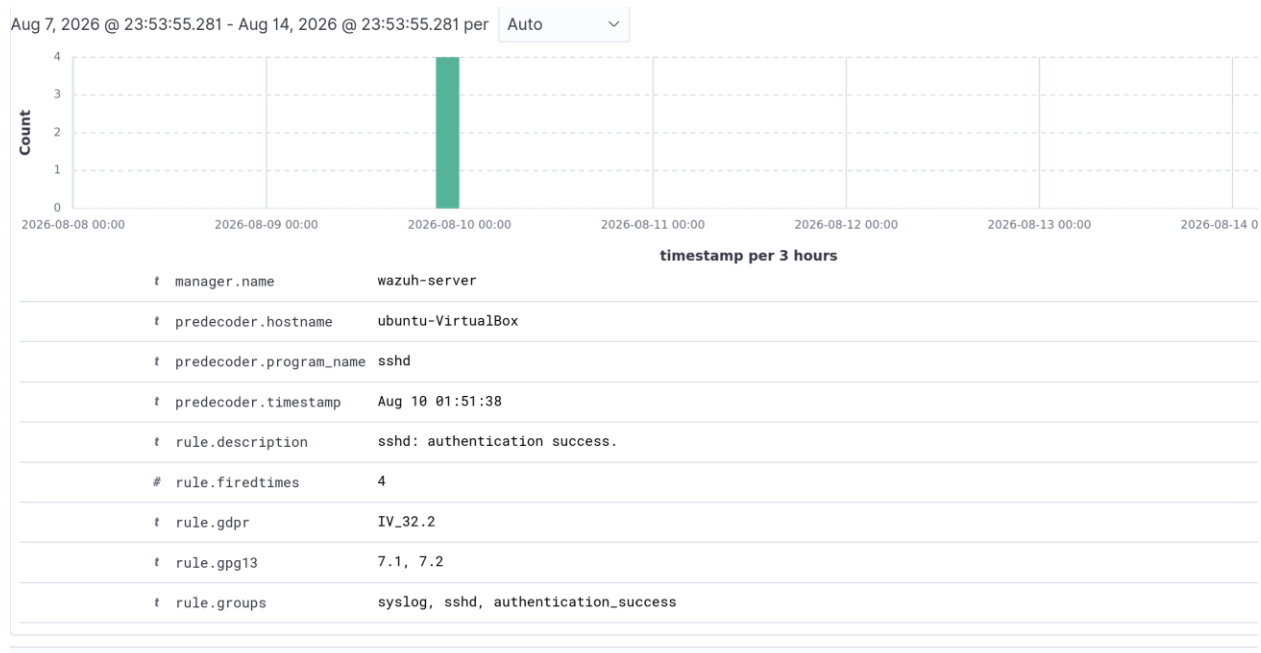


f rule.gpg13	7.8
f rule.groups	pam, syslog, authentication_failures
f rule.hipaa	164.312.b
f rule.id	5551
# rule.level	10
rule.mail	false
f rule.mitre.id	T1110
f rule.mitre.tactic	Credential Access
f rule.mitre.technique	Brute Force
f rule.nist_800_53	AU.14, AC.7, SI.4



Aug 7, 2026 @ 23:24:47.762 - Aug 14, 2026 @ 23:24:47.763 per





- Port scanning or reconnaissance alerts
- File Integrity Monitoring (FIM) alerts (if configured)

Document Details

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Table JSON

t _index	wazuh-alerts-4.x-2026.08.09
t agent.id	001
t agent.ip	192.168.30.11
t agent.name	MedCore
t decoder.name	syscheck_integrity_changed
t full_log	File '/usr/bin/acpidbg' modified Mode: scheduled Changed attributes: inode,mtime Old modification time was: '1782935281', now it is '1784295868' Old inode was: '1314459', now it is '1314676'
t id	1786239188.3017
t input.type	log
t location	syscheck
t manager.name	wazuh-server
t rule.description	Integrity checksum changed.
# rule.firedtimes	3
t rule.gdpr	II_5.1.f
t rule.gpg13	4.11
t rule.groups	ossec, syscheck, syscheck_entry_modified, syscheck_file
* rule.hmac	164 312 c 1 164 312 c 2

- Active Response or blocked IP events

```
Session Actions Edit View Help
(kali㉿kali)-[~]
$ ping 192.168.30.10
PING 192.168.30.10 (192.168.30.10) 56(84) bytes of data.
^C
— 192.168.30.10 ping statistics —
27 packets transmitted, 0 received, 100% packet loss, time 26679ms

— 192.168.30.10 ping statistics —
0 packets transmitted, 0 received, 0% packet loss, time 3001ms
679/9.597/13.343/2.298 ms
(kali㉿kali)-[~]
$ ping 192.168.30.11
PING 192.168.30.11 (192.168.30.11) 56(84) bytes of data.
^C
— 192.168.30.11 ping statistics —
10 packets transmitted, 0 received, 100% packet loss, time 9315ms

(kali㉿kali)-[~]
$
```

CHAPTER TEN

10.0 Key Analytical Questions

Network Design

1. Which assets belong in the LAN, WAN, and DMZ zones?

The LAN contains internal systems such as clinical and administrative workstations, SOC workstations, Wazuh SIEM, IoMT devices, and the Kali Linux workstation used for controlled attack simulation. The DMZ contains public-facing services, including the EHR server, Patient Portal, and Telehealth Server. The WAN represents the external or Internet-facing network. This segmentation separates internal resources from systems exposed to external traffic and reduces unauthorized access to sensitive healthcare systems.

Firewall Security

2. Which traffic should be allowed between zones?

Only business-required traffic should be permitted between the network zones(e.g an admin account only which is configured as aliases in pfsense). HTTPS traffic on TCP port 443 should be allowed from the Internet to the Patient Portal in the DMZ. Authorized clinical workstations should be permitted to access the EHR server, Wazuh agents should communicate with the Wazuh Manager, and required DNS and DHCP traffic should be permitted. All access should follow the principle of least privilege.

3. Which traffic should be blocked?

Unnecessary or unauthorized traffic between network zones should be blocked. Direct Internet access to internal systems should not be permitted. IoMT devices should be prevented from initiating connections to administrative systems, and unsolicited inbound traffic should be denied unless specifically authorized. Suspicious or malicious traffic, including repeated unauthorized SSH connection attempts, should also be blocked.

Threat Detection

4. How can suspicious internal traffic be identified?

Suspicious internal traffic can be identified by combining Wazuh SIEM monitoring, pfSense firewall logs, and Wireshark packet analysis. Indicators include repeated TCP SYN packets, port scanning, multiple failed SSH authentication attempts, unusually high numbers of connections from one IP address, firewall deny events, and unauthorized changes to critical systems.

Correlating timestamps, source and destination IP addresses, ports, usernames, and event severity helps the SOC distinguish normal activity from malicious behaviour.

Risk Assessment

5. What business risks exist if segmentation is not implemented?

Without proper network segmentation, an attacker who compromises one system may be able to move laterally (lateral movement) to other critical systems. This increases the risk of unauthorized access to EHRs, exposure of Protected Health Information (PHI), modification of clinical information, service disruption, and compromise of connected medical devices. These incidents could affect patient care, cause system downtime, result in regulatory non-compliance, and damage MediCore Health Systems' reputation.

Security Improvement

6. What additional controls should be introduced to strengthen security?

Additional controls should include Multi-Factor Authentication (MFA), SSH hardening, key-based authentication, stricter default-deny firewall policies, File Integrity Monitoring (FIM), centralized log retention, and improved Wazuh SIEM detection rules. MediCore should also conduct regular vulnerability assessments and penetration testing, implement patch management, provide cybersecurity awareness training, and adopt Zero Trust principles. These controls would strengthen prevention, detection, and response while improving protection of sensitive healthcare information.